15th Sep 2026

1 1.008	92 238.03	9 19.00	16 32.06
H Hydrogen	**U** Uranium	**F** Fluorine	**S** Sulfur

재생에너지 기술 및 응용

(Renewable Energy Technologies and Applications)

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Hydrogen **E**nerg**Y** Lab **ON** **E**arth

Assistant Professor

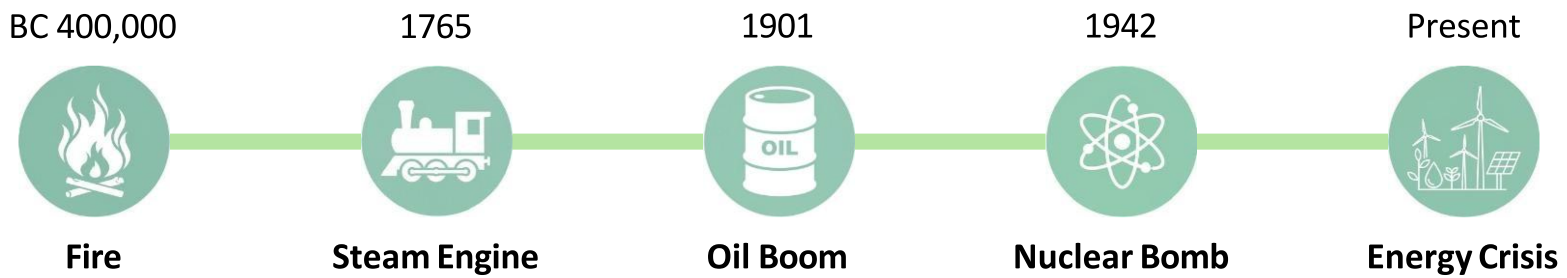
Division of Climate Change

Hankuk University of Foreign Studies (HUFS)

Review

Timeline of Energy

❑ Energy Transition



Sir Benjamin Thompson (1753–1814)	*The Nature of Heat (1798)* "Heat is a form of energy corresponding to a definite amount of mechanical work."
Julius Robert von Mayer (1814–1878)	*The Laws of Conservation of Energy (1842)* "Energy can be neither created nor destroyed"
William John Macquorn Rankine (1820–1872)	*Potential Energy (1853)*
William Thomson, Lord Kelvin (1824–1907)	*Kinetic Energy (1867)*
Albert Einstein (1879–1955)	*The Theory of Relativity (1905)* "Mass and energy are equivalent and interchangeable"

Types of Energy

❑ Nonrenewable Energy

Energy derived from natural resources that cannot be readily replenished in a short period

*Millions of years to be formed

Oil

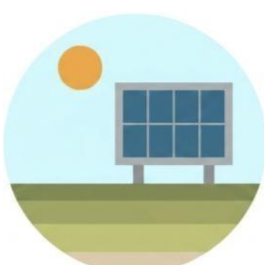
Coal

Natural Gas

Nuclear

❑ Renewable Energy

Energy derived from natural resources that are replenished on a human timescale

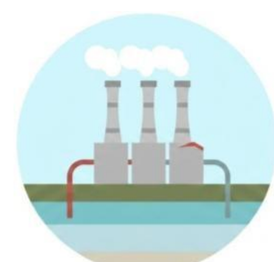


Solar

Wind

Hydropower

Biomass

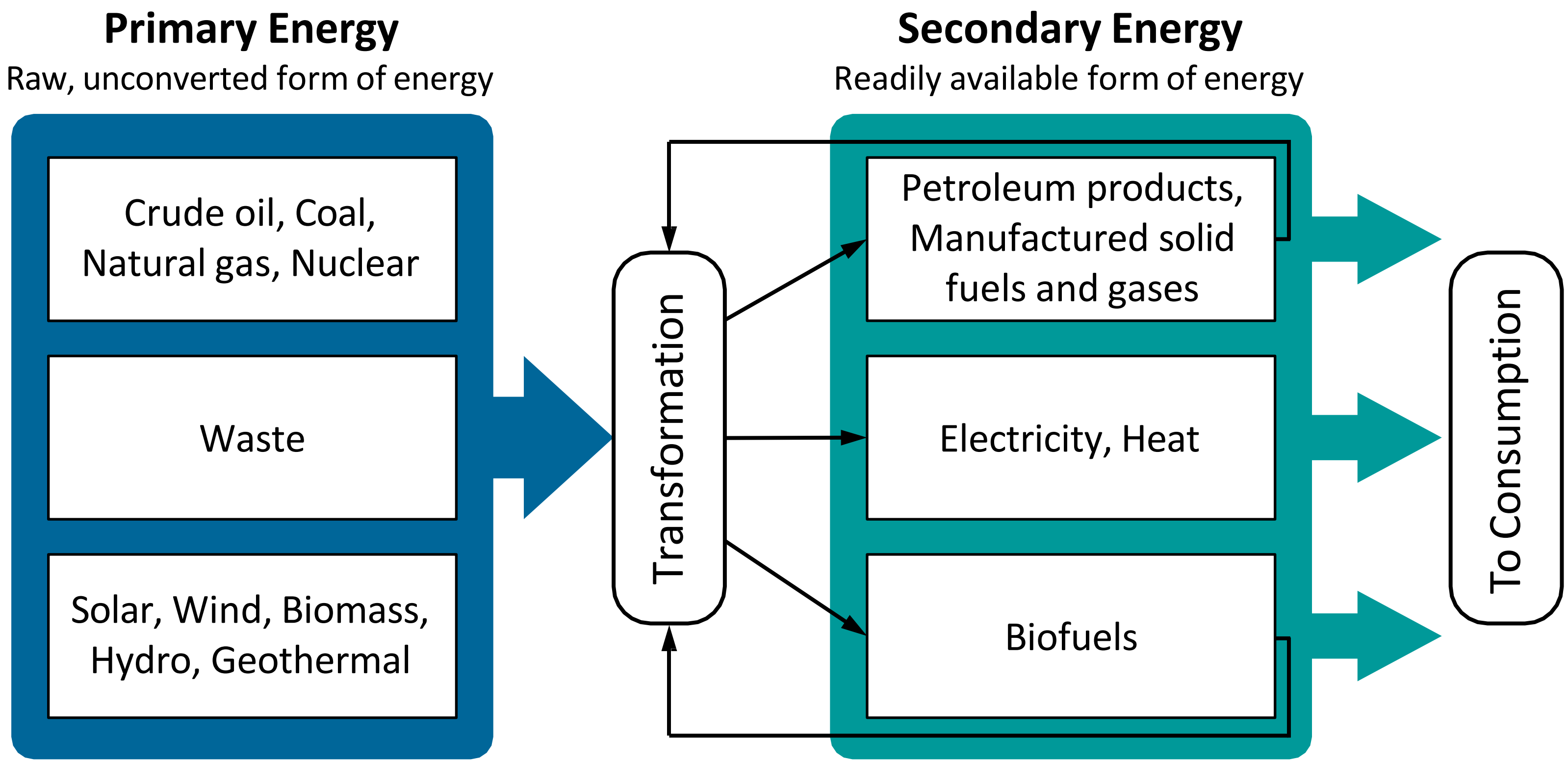


Geothermal Heat

Types of Energy

❑ Primary and Secondary Energy

Energy is indeed abundant, but *usable* energy is strictly limited, making *tools* essential



– *Usable* energy must be transformed from primary energy sources

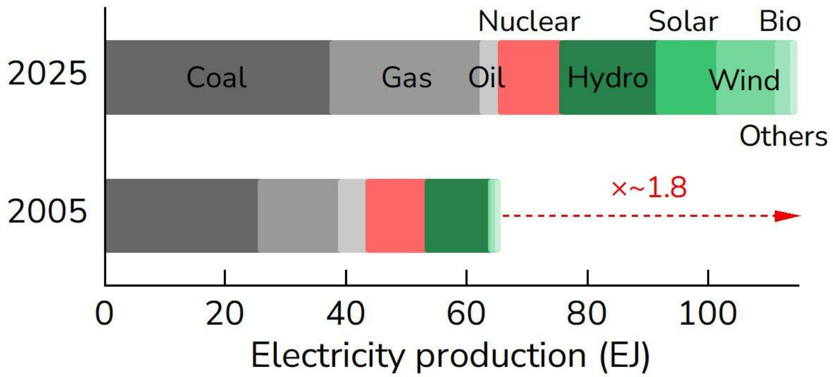
– Every energy conversion process must entail *losses*

Overview of Global Electricity

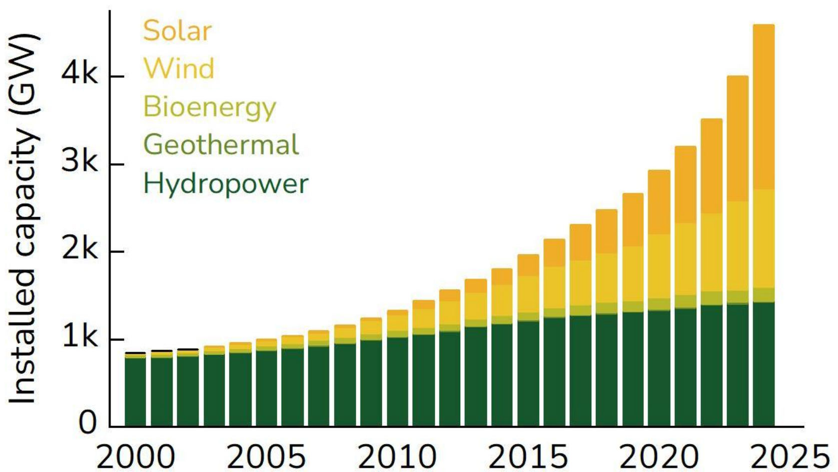
❑ Global Electricity Production

RE accounts for 34% in 2025 (18% in 2005)

*Others: geothermal, biomass, waste, wave and tidal



<Source: Statistical Review of World Energy (2025)>

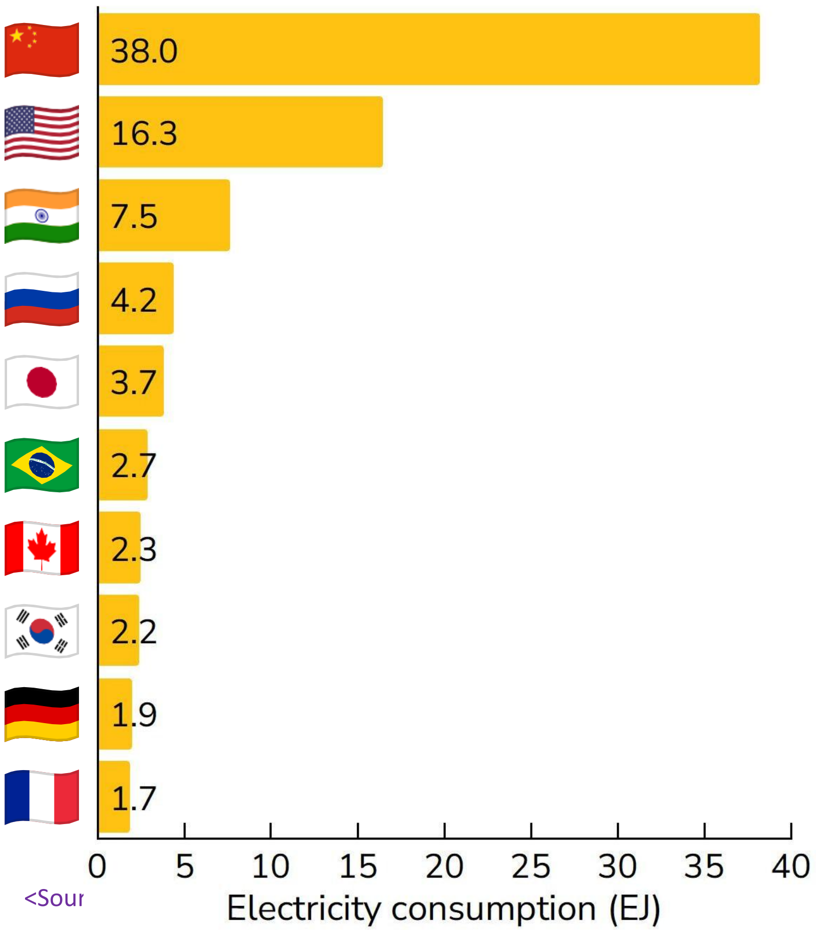


<Source: IRENA (2025)>

❑ Global Electricity Demand

Data centers consumed ~1.8 EJ in 2025

**Usage of ~3.4 EJ in 2030

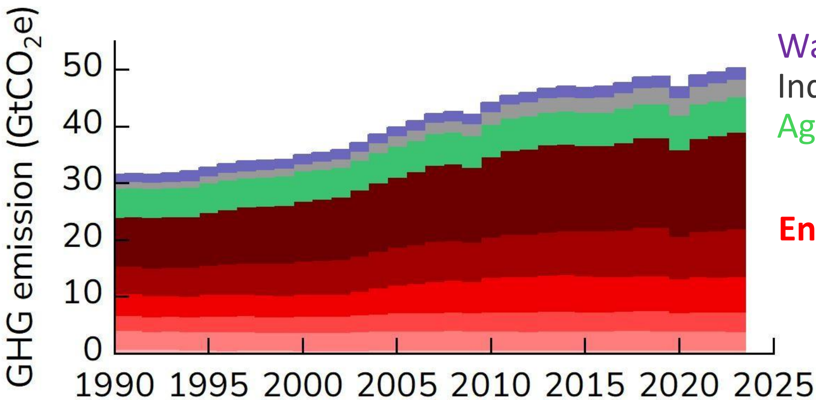


<Source: Ember (2026), Energy and AI (2026)>

Why Renewables?

❑ Global GHG Emissions by Sector

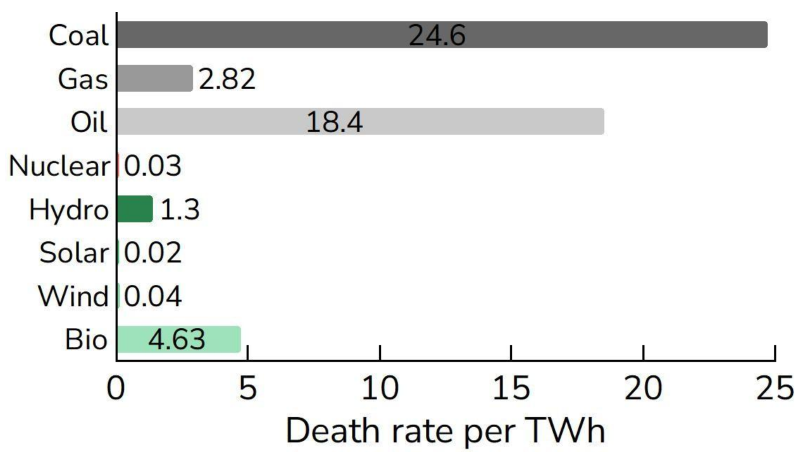
Greenhouse gas emissions in 2023 were ~50 billion tonnes CO_2eq, with ~80% coming from *energy*



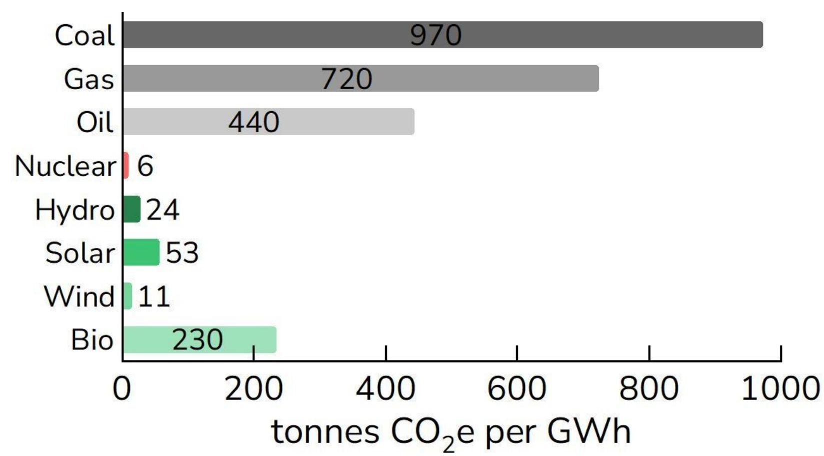
Waste (3.9%) (Wastewater, Landfill, etc.)
Industry (6.3%) (e.g., Cement, Chemical, etc.)
Agriculture, Forestry (12%) (e.g., Livestock, Deforestation, etc.)

Energy (77%) (e.g., Electricity, Heat, Transport, etc.)

❑ Air Pollution & Accident



❑ GHG Emissions

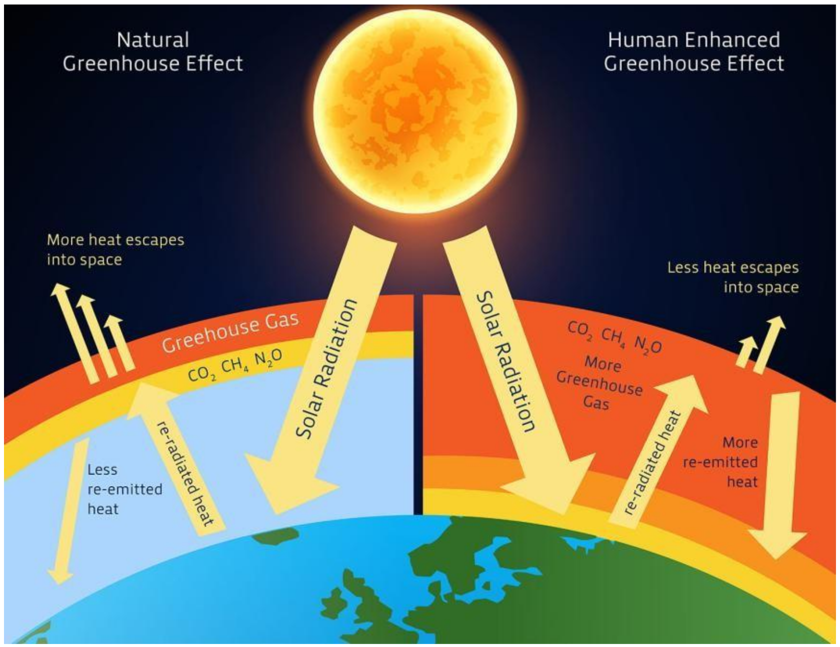


<Source: Our World in Data>

Greenhouse Effect

❑ Greenhouse Effect

GHGs prevent the planet from losing heat to space



Renew. Sustain. Energy Rev., **71** (2017) 112–126

❑ Earth's Energy Budget

Annual Earth Energy Imbalance = ~0.6 W m^{-2}

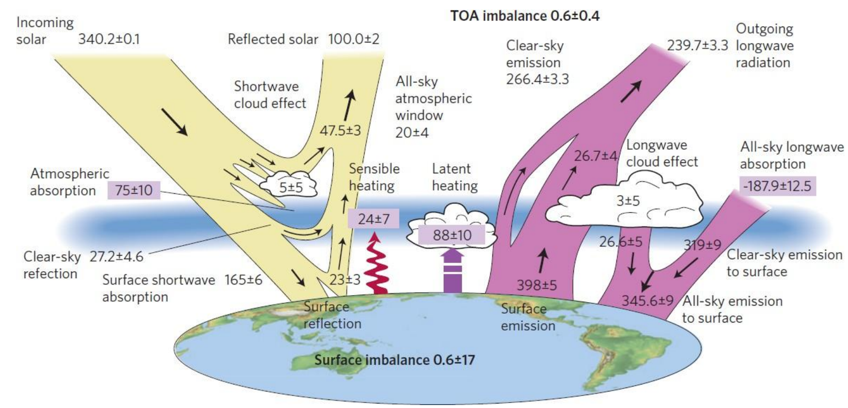


Fig. Global annual mean energy budget of Earth for the approximate period 2000–2010.

Yellow: Solar fluxes
Pink: Infrared fluxes

Nat. Geosci., **5** (2012) 691–696

– The surface of the Sun is ~5500 °C, so it emits *shortwave* radiation (UV, VL, IR)

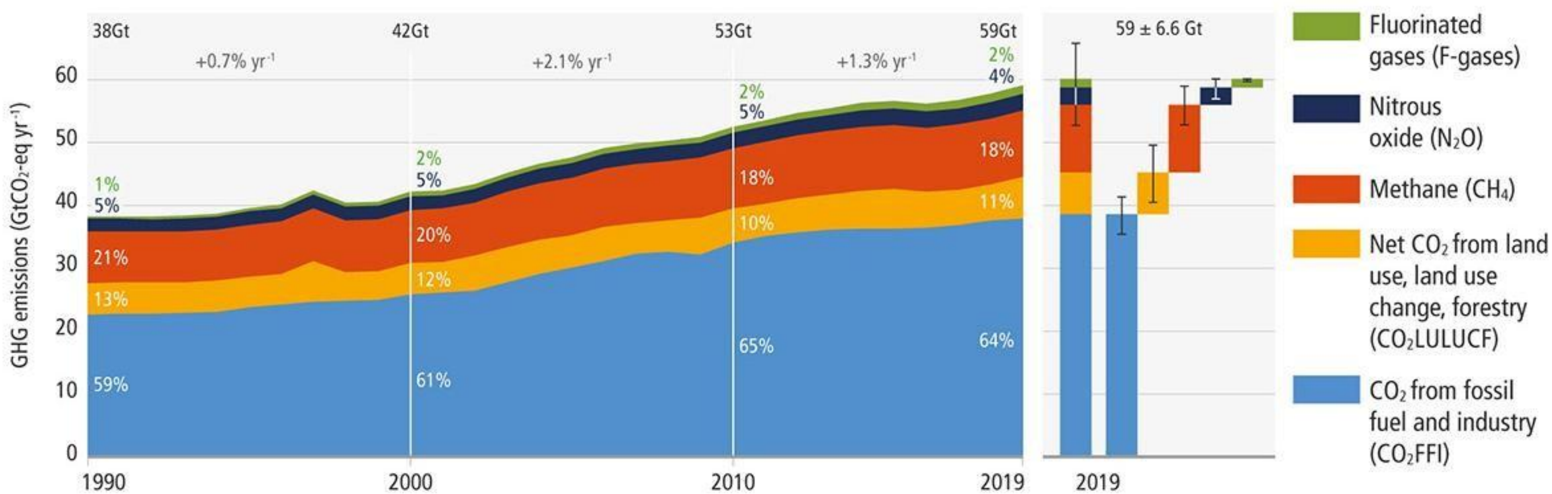
– Shortwave radiation passes through GHGs and heats the Earth's surface

– The Earth's surface emits *longwave* radiation that is mostly absorbed by GHGs

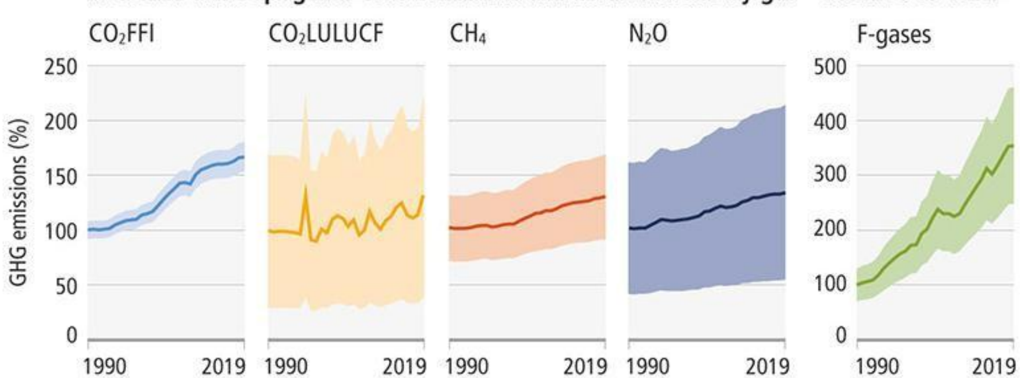
Greenhouse Effect

❑ Global Net Anthropogenic GHG Emissions

a. Global net anthropogenic GHG emissions 1990–2019 [5]



b. Global anthropogenic GHG emissions and uncertainties by gas – relative to 1990



	2019 emissions ($GtCO_2$-eq)	1990–2019 increase ($GtCO_2$-eq)	Emissions in 2019, relative to 1990 (%)
CO_2 FFI	38±3	15	167
CO_2 LULUCF	6.6±4.6	1.6	133
CH_4	11±3.2	2.4	129
N_2O	2.7±1.6	0.65	133
F-gases	1.4±0.41	0.97	354
Total	59±6.6	21	154

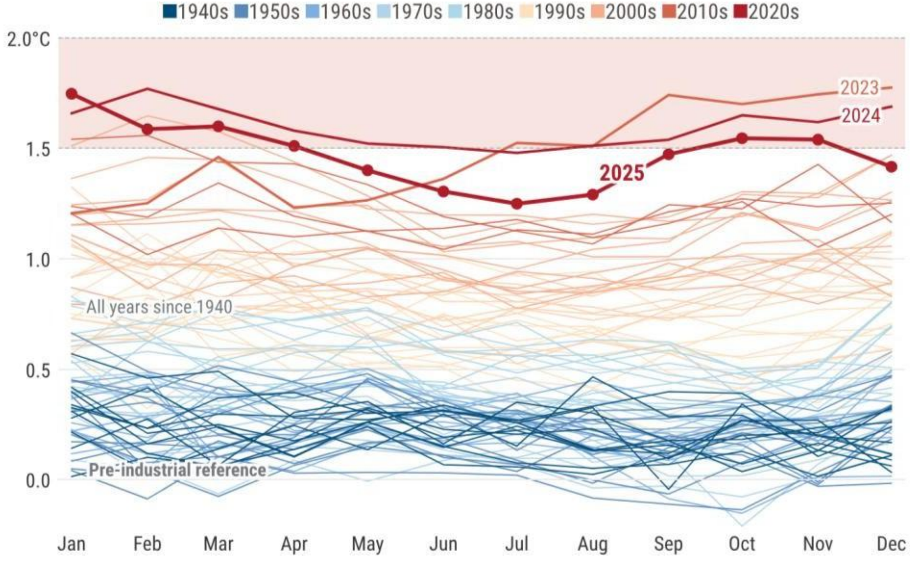
The solid line indicates central estimate of emissions trends. The shaded area indicates the uncertainty range.

<Source: IPCC>

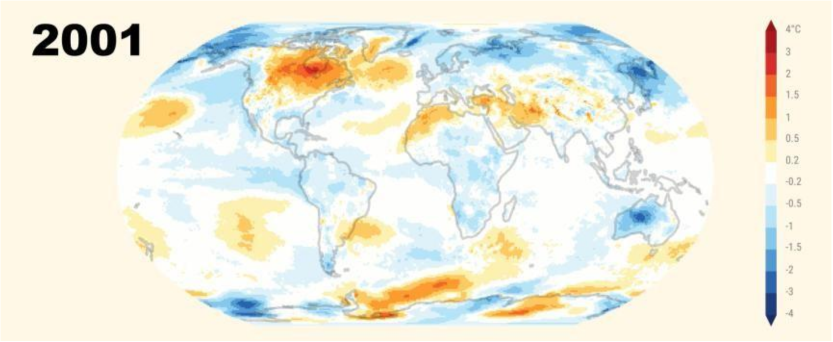
Climate Change

❑ Global Boiling

Monthly global surface air temperature anomalies



Annual surface air temperature anomalies (2001–2025)



<Source: Corpernicus Climate Change (2025)>

Climate Change-Driven Disasters

❑ Heatwave

Hajj extreme heat disaster (Jun 2024)

❑ Tropical Cyclone

Hurricane Helene (Sep 2024)

❑ Wildfire

LA wildfire (Jan 2025)

❑ Drought

Amazon deforestation

❑ Flood

Spanish floods (Oct–Nov 2024)

❑ Particulate matter

Air pollution in London

Climate Impacts

❑ Temperature Rise Scenarios and Climate Change Impacts

	1.5°C	2°C	3°C	1.5°C vs 2°C	1.5°C vs 3°C
BIODIVERSITY LOSS Maximum percentage of species at high risk of extinction across forests and land	14%	18%	29%	1.3x WORSE	2.1x WORSE
DROUGHT Dryland population exposed to water stress, heat stress and desertification	0.95B PEOPLE	1.15B PEOPLE	1.29B PEOPLE	200M MORE PEOPLE	340M MORE PEOPLE
FOOD SECURITY Costs for adaptation and residual damage to major crops	$63 BILLION US	$80 BILLION US	$128 BILLION US	$17B MORE	$65B MORE
FIRES Increases in burnt area across Mediterranean Europe	40-54%	62-87%	96-187%	1.6x WORSE	3x WORSE
EXTREME HEAT Increase in number of days per year with a maximum temperature above 35°C (95°F)	45-58	52-68	66-87	1.2x WORSE	1.5x WORSE
EXTREME HEAT Increase in annual number of heatwaves in Southern Africa	2-4 TIMES	4-8 TIMES	8-12 TIMES	2x WORSE	3.3x WORSE

Climate Impacts

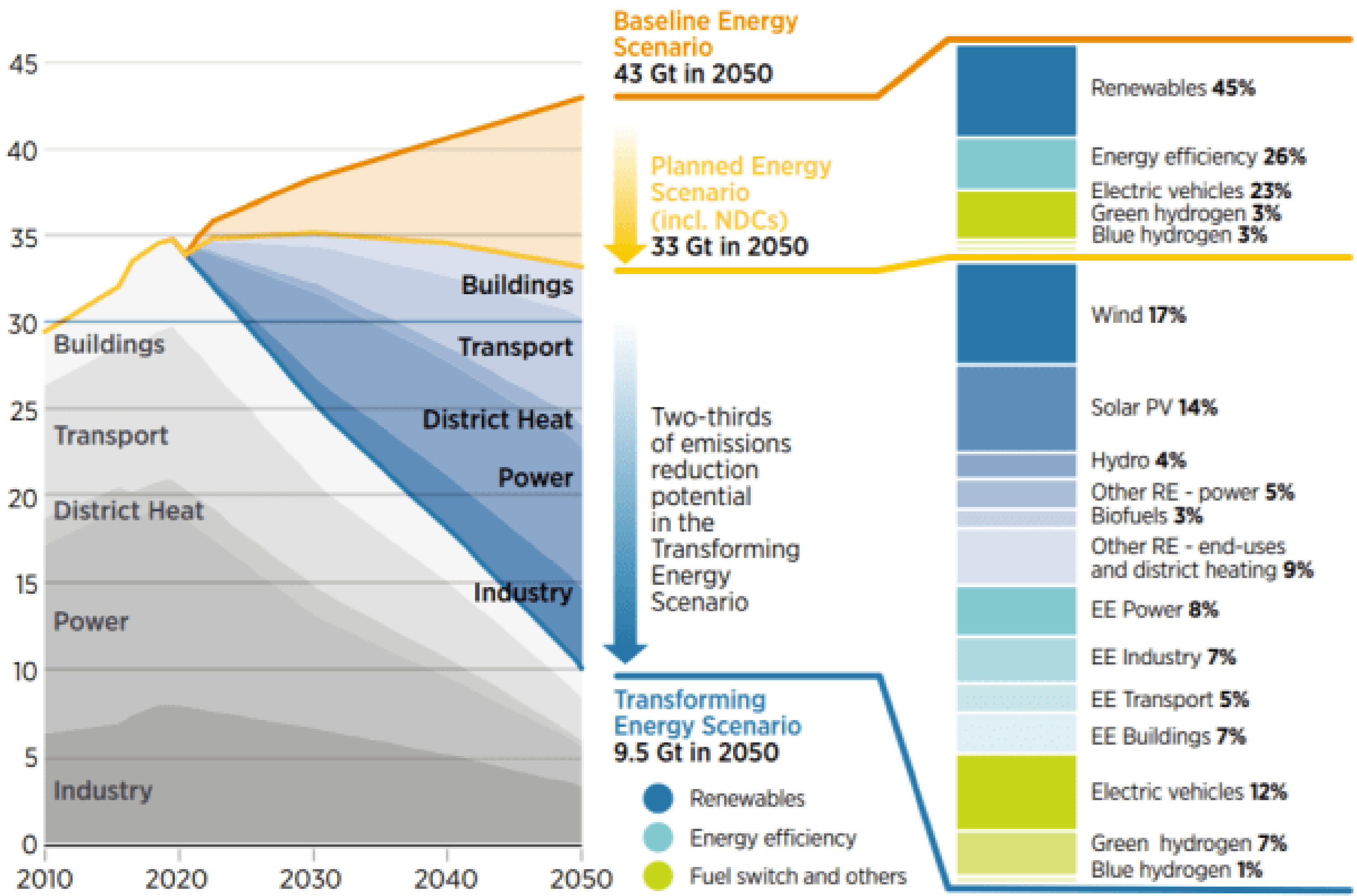
❑ Temperature Rise Scenarios and Climate Change Impacts (Cont'd)

	1.5°C	2°C	3°C	1.5°C vs 2°C	1.5°C vs 3°C
SEA LEVEL RISE Global mean sea level rise by 2100	0.28-0.55m	0.33-0.61m	0.44-0.76m	1.1x WORSE	1.4x WORSE
FLOODS Increase in global population exposed to flooding	24%	30%	NO DATA AVAILABLE	1.3x WORSE	NO DATA AVAILABLE
CORAL REEFS Further decline in coral reefs	70-90%	99%	NO DATA AVAILABLE	1.2x WORSE	NO DATA AVAILABLE

<Source: Climate Resources Institute>

Pathways to 1.5 °C

❑ Mitigating GHG Emissions by Energy Transition



<Source: GLOBAL ENERGY TRANSFORMATION - a roadmap to 2050>

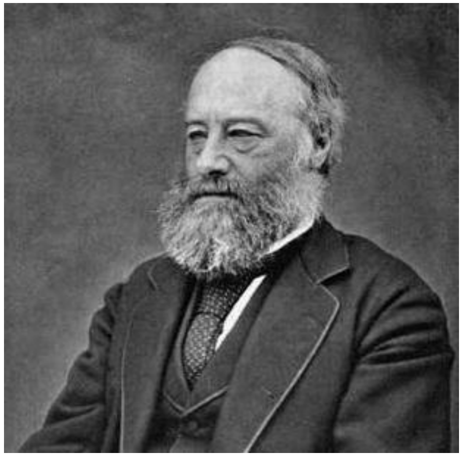
2. Fundamentals of Thermodynamics

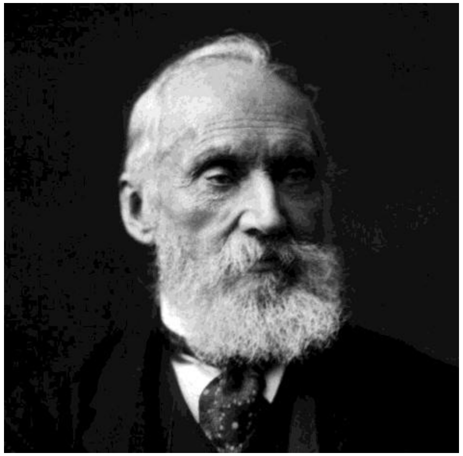
Contents

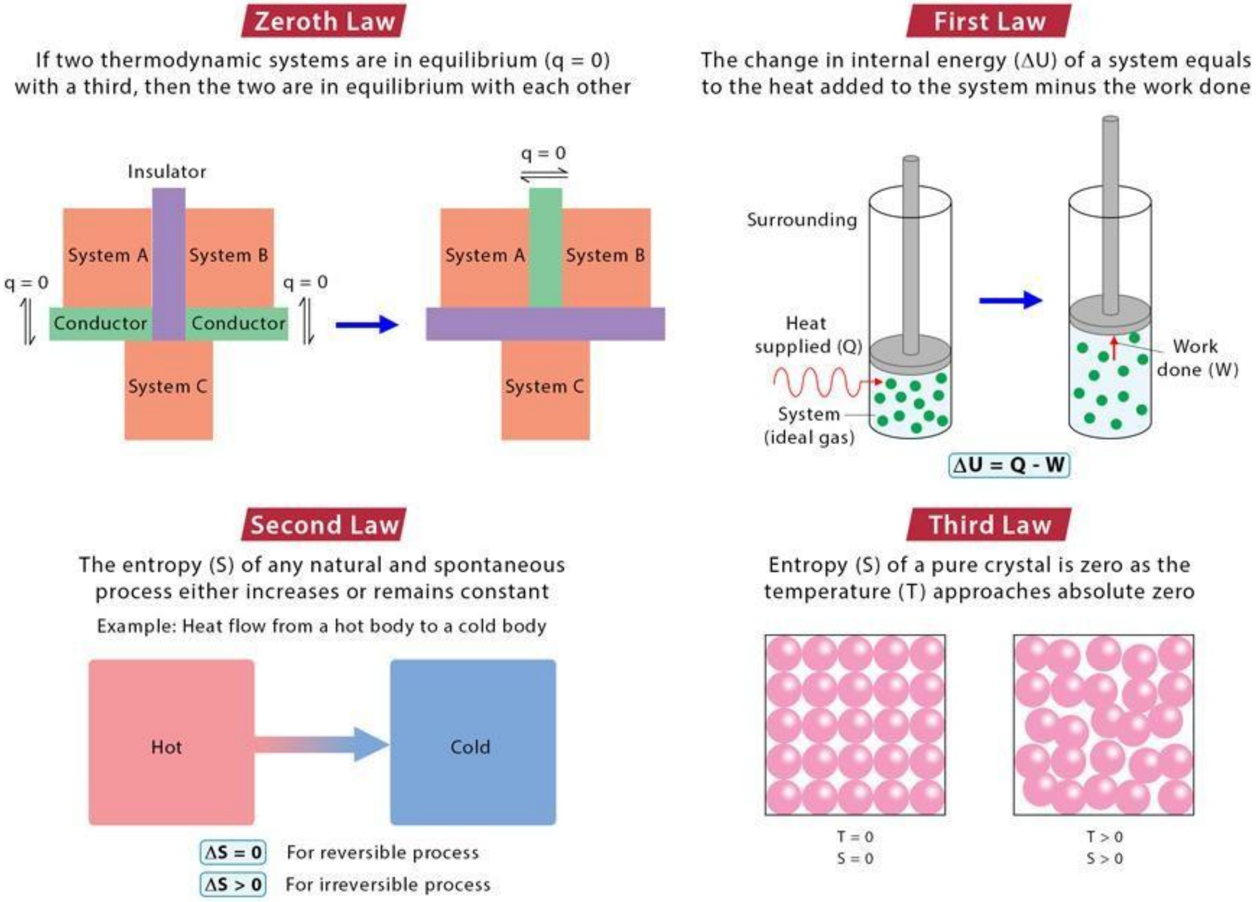
What is Thermodynamics?

❑ Thermodynamics

A unified theory that bridges thermotics (temperature, heat) and mechanics (work, pressure)



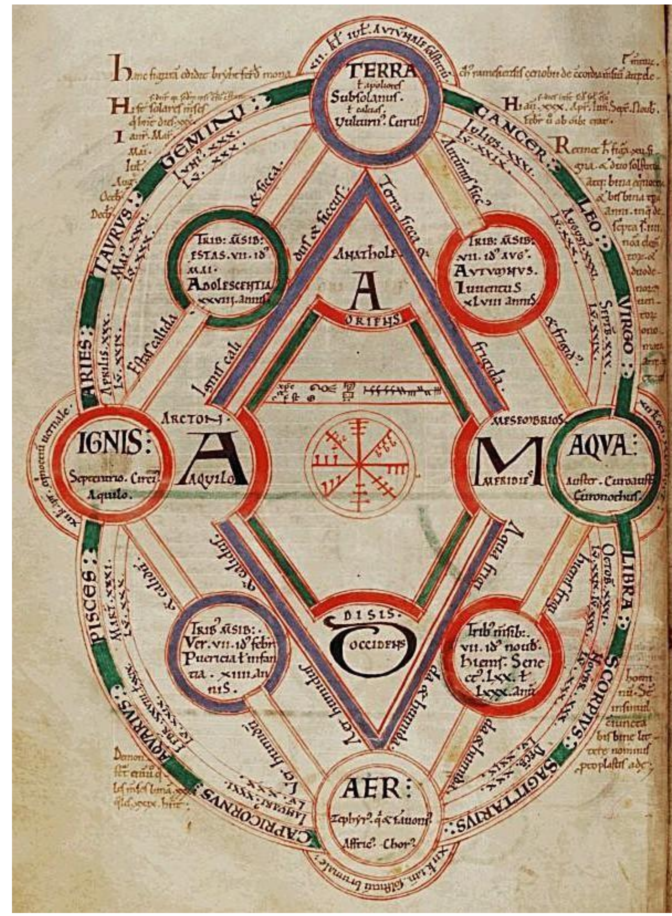




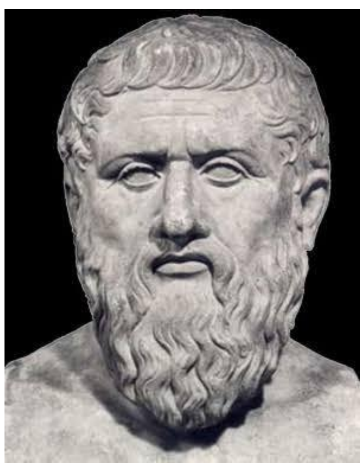
What is Thermodynamics?

❑ Four-Element Theory

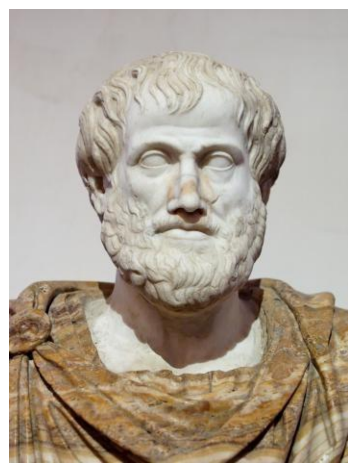
All matter in the universe is made of four basic components: earth, **fire**, water, and air



Empedocles
(BC 494–BC 434)



Platon
(BC 428–BC 348)

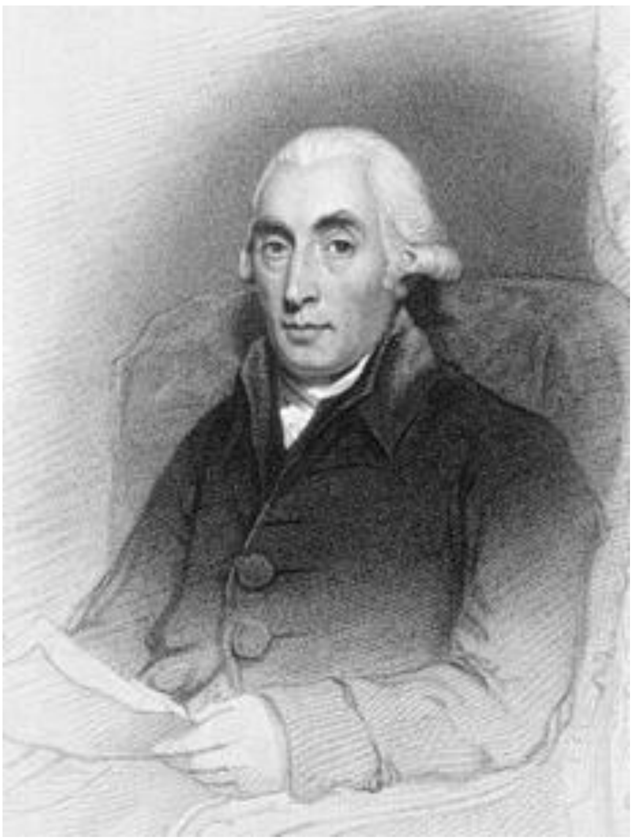


Aristotle
(BC 384–BC 322)

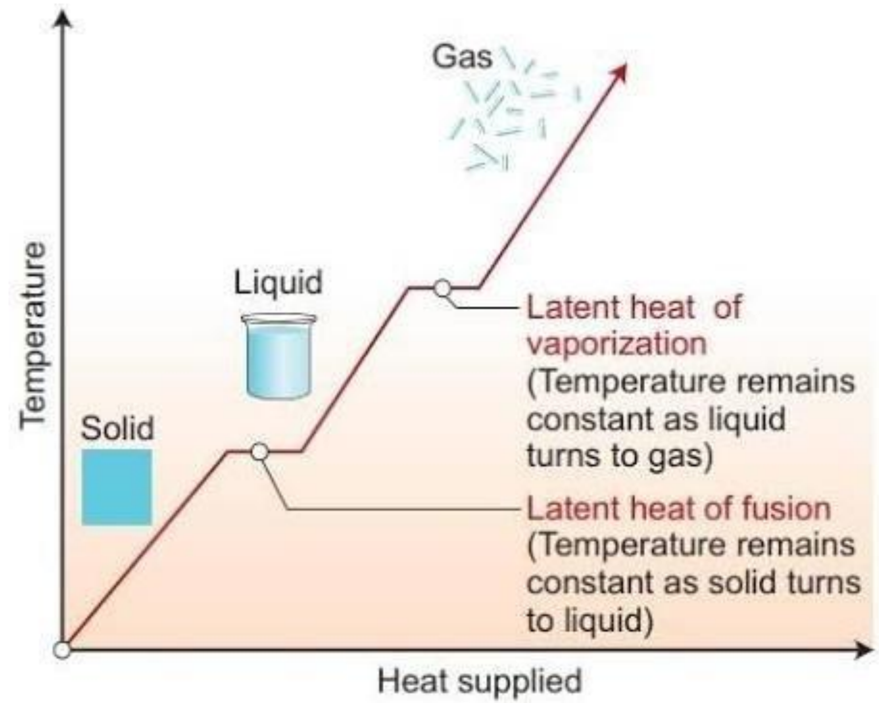
What is Thermodynamics?

❑ Caloric Theory (superseded scientific theory)

Heat consists of a self-repellent fluid called “caloric” that flows from hotter bodies to colder bodies



Joseph Black
(1728–1799)



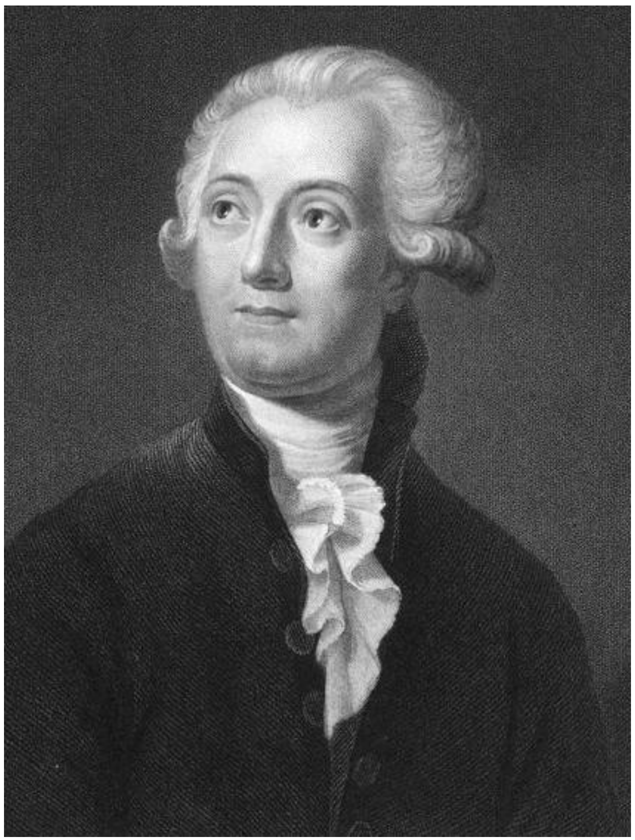
“Discovery of Latent Heat” (1761)

Applying heat to ice at its melting point does not cause a rise in temperature of the ice/water mixture, but rather an increase in the amount of water in the mixture.

What is Thermodynamics?

❑ Caloric Theory (superseded scientific theory)

Heat consists of a self-repellent fluid called “caloric” that flows from hotter bodies to colder bodies



Antoine-Laurent de Lavoisier
(1743–1794)

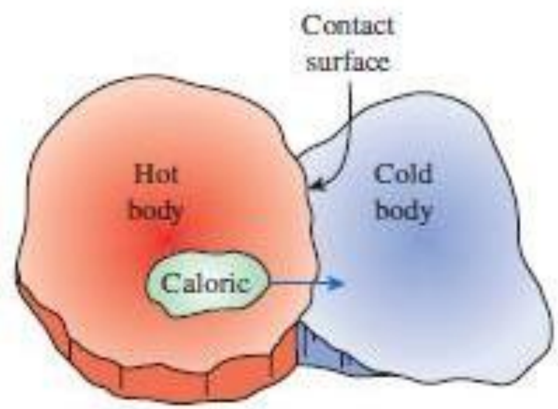


FIGURE 2–19
In the early 19th century, heat was thought to be an invisible fluid called the *caloric* that flowed from warmer bodies to cooler ones.

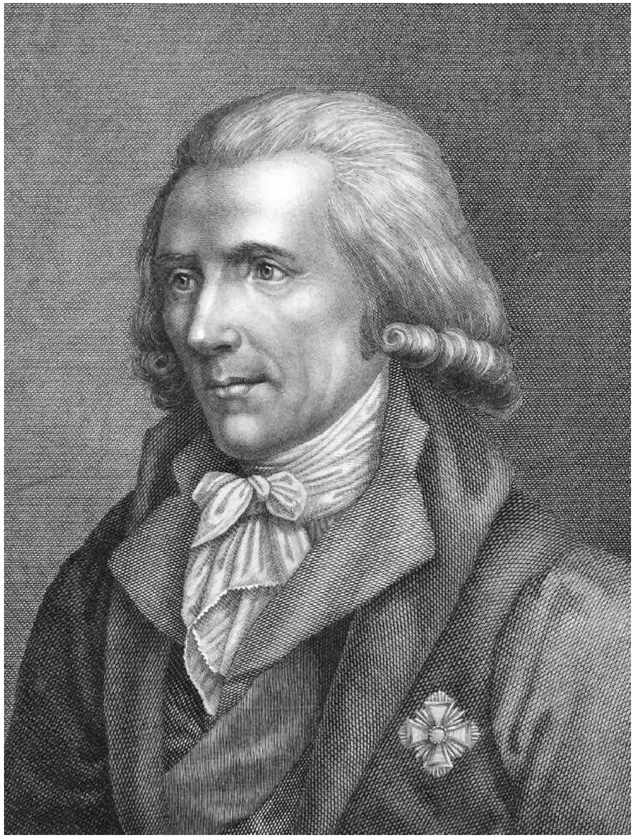
“Method of Chemical Nomenclature” (1787)

Elements are defined as substances that could not be decomposed into simpler substances by any known chemical means, including light and ***caloric (matter of heat)***

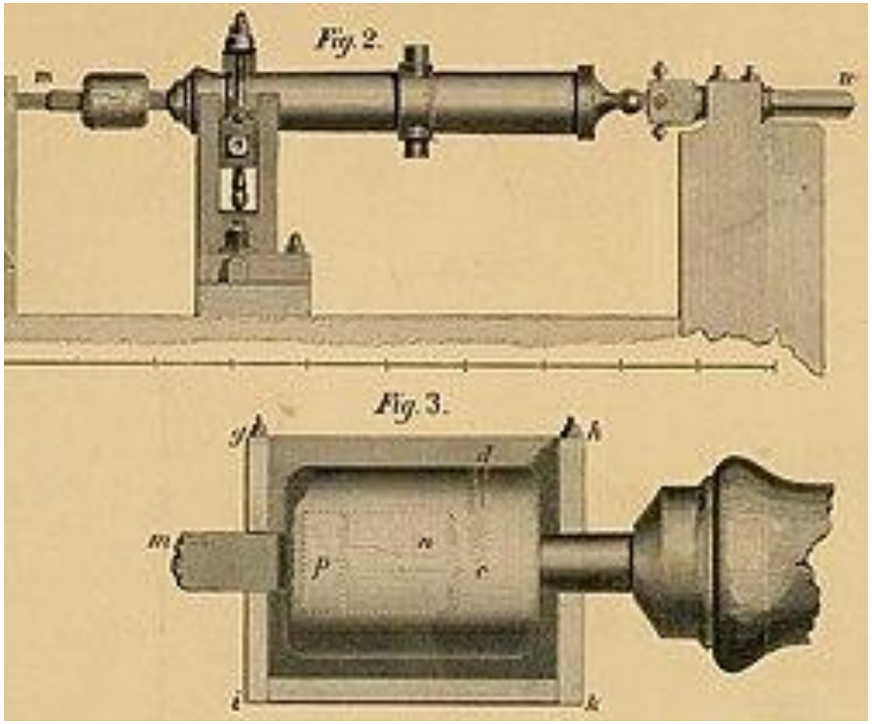
What is Thermodynamics?

❑ Rumford's Theory of Heat

Heat is not a material fluid (caloric), but a form of motion



Benjamin Thompson, Count Rumford
(1753–1814)



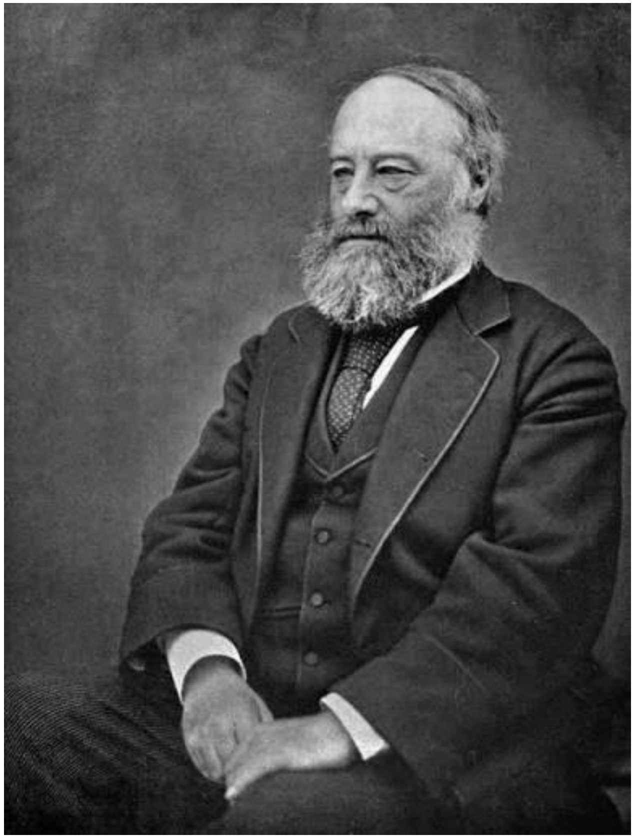
"An Inquiry Concerning the Source of the Heat Which Is Excited by Friction" (1798)

Observing the boring of cannon barrels revealed continuous, inexhaustible heat generated by friction

What is Thermodynamics?

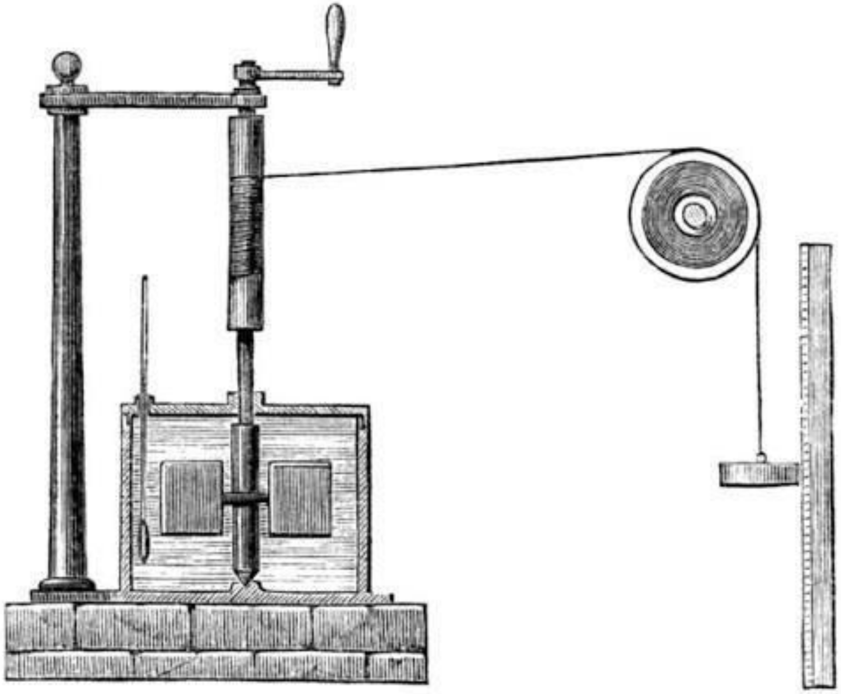
❑ Mechanical Equivalent of Heat

Motion and heat are mutually interchangeable and that in every case, a given amount of work would generate the same amount of heat, provided the work done is totally converted to heat energy



James Prescott Joule
(1818–1889)

$$C\Delta T = mg\Delta h$$



"On the Mechanical Equivalent of Heat" (1849)

The quantity of heat produced by the friction of bodies is always proportional to the quantity of force expended

What is Thermodynamics?

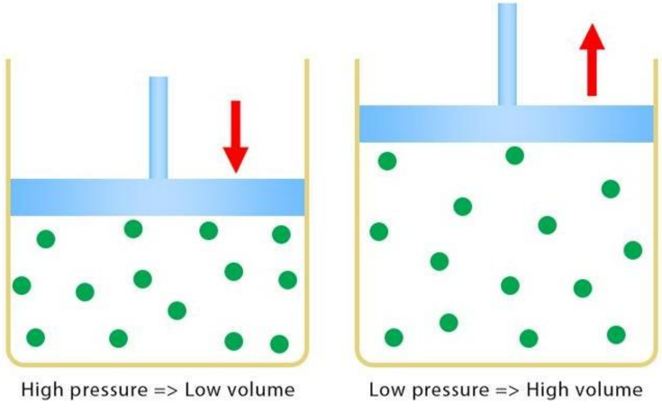
❑ Thermodynamics

A unified theory that bridges thermotics (temperature, heat) and mechanics (work, pressure)

- **Characteristic Examples**
  (1) Measuring one physical quantity allows us to predict another

Boyle's Law

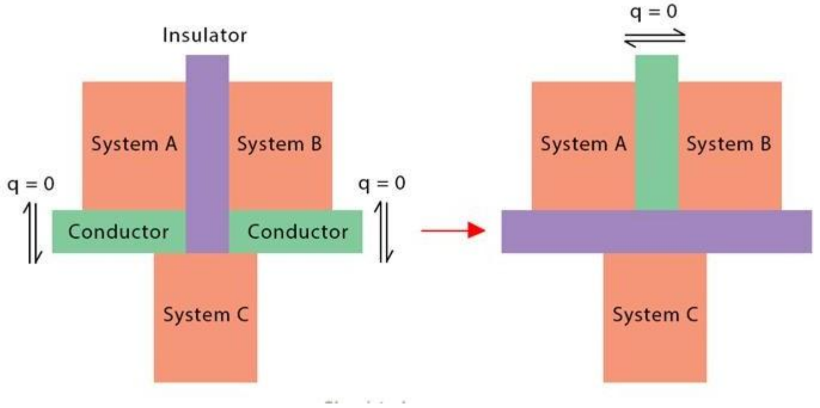
The pressure and volume of a gas are inversely proportional, provided the temperature and mass are kept constant



(e.g.) Consider an ideal gas at a constant temperature of 300 K.
Initial state: p_i = 2 atm, V_i = 2 L
Final state: p_f = 1 atm, V_f = ? L

0^{th} Law of Thermodynamics

If two thermodynamic systems are in equilibrium (q = 0) with a third, then the two are in equilibrium with each other



(e.g.) System A and System B are in thermal equilibrium with System C.
Estimate T_C when T_A = T_B = 300 K
T_c = ? K

What is Thermodynamics?

Thermodynamics

A unified theory that bridges thermotics (temperature, heat) and mechanics (work, pressure)

- **Characteristic Examples**

  (2) The volume dependence of the heat capacity C_V at constant volume V is given by *Maxwell relation*

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T\left(\frac{\partial^2 P}{\partial T^2}\right)_V$$

(e.g.) The van der Waals equation can be written as

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT \Leftrightarrow P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

where *p* is pressure, *R* is the gas constant, *T* is temperature, and *V* is volume, and *n* is the number of moles in the gas. The constants *a* and *b* are experimentally determinable, substance-specific constants.

First derivative with respect to T: $\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V - nb}$

Second derivative with respect to T: $\left(\frac{\partial^2 P}{\partial T^2}\right)_V = 0$

Volume dependence result: $\left(\frac{\partial C_V}{\partial V}\right)_T = T\left(\frac{\partial^2 P}{\partial T^2}\right)_V = 0$

What Thermodynamics Deals With

❑ Material-Specific Properties vs. Universal Laws

Thermodynamics strictly separates material-specific properties from universal laws

- **Material-Specific Properties**

  A substance is completely characterized by volume V and temperature T, where the quantities depend on the substance species A and the amount of substances N

  (e.g.) Heat capacity $C = C(T,V;\ A,N)$
  Equation of state $P = P(T,V;\ A,N)$

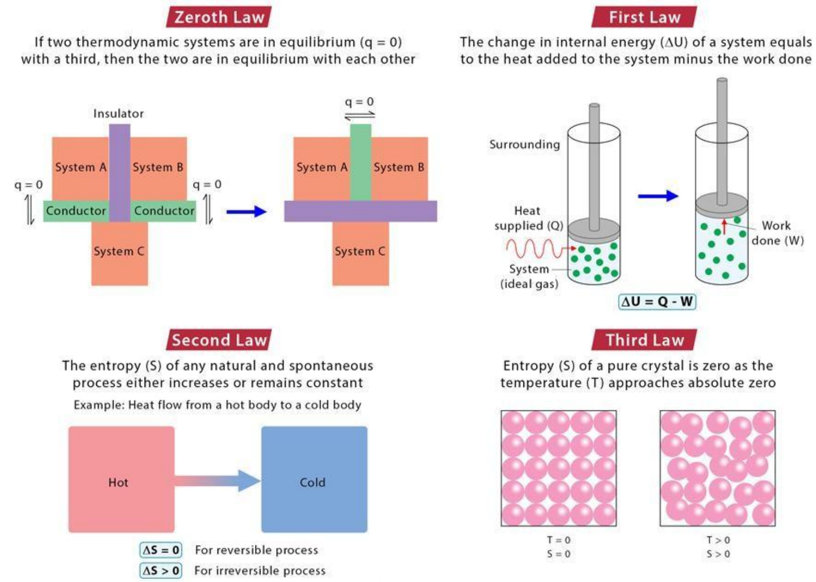
Simple Substance

Volume (V)
Temperature (T)

- **Thermodynamic Laws**

  Universal relations governing any macroscopic system regardless of material type

  (e.g.) Zeroth Law: A~B ∧ B~C ⇒ A~C
  First Law: $\Delta U = Q - W$
  Second Law: $\delta Q = TdS$
  Third Law: $T \rightarrow 0$ K, $S \rightarrow S_0$ (constant)



What Thermodynamics Deals With

❑ Aims & Scope

The primary goal is establishing relationships between material properties using thermodynamic laws

- **Primary Goal**

  Use the laws of thermodynamics to discover, derive, and constrain the exact relationships governing the individual properties of matter.

  The essential milestones include:

  (1) Comprehend and apply the fundamental laws of thermodynamics
  (2) Categorize material-specific properties and ***macroscopic*** *phenomena*
  (3) Familiarize with new abstract concepts, such as ***internal energy, entropy, free energy, etc.***

What Thermodynamics Deals With

❑ Macroscopic vs. Microscopic

Thermodynamic targets must be divisible without altering their fundamental thermodynamic identity

- **The "Biscuit Principle"**
  When a macroscopic body is partitioned into two halves, each divided subsystem retains the same thermodynamic state as the original

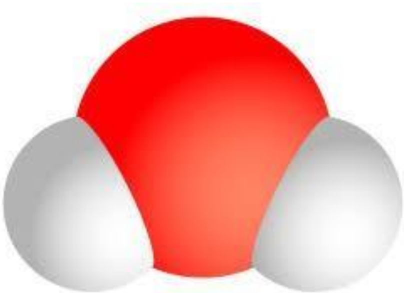
Macroscopic Level (Water)

Splitting water in half yields two identical samples of water

Thermodynamics deals with ***macroscopic matter***

Microscopic Level (Molecule)

Splitting an H_2O molecule in half breaks chemical bonds; it is no longer water.



Operational Definition

❑ Operational Definition

All thermodynamic concepts must be defined strictly through executable experimental procedures

- **Ambiguity of Terminology**

  Everyday words like *heat* are often ambiguous and misleading.

  Physical quantities must be anchored in **operational definitions**, not intuition

- **A quantity is defined by the exact experimental steps used to measure it**

  An empirical science can also be verified by third-party exp., unlike modern mathematics

  *Axiom

Thermometer

Manometer

Weighing Scale

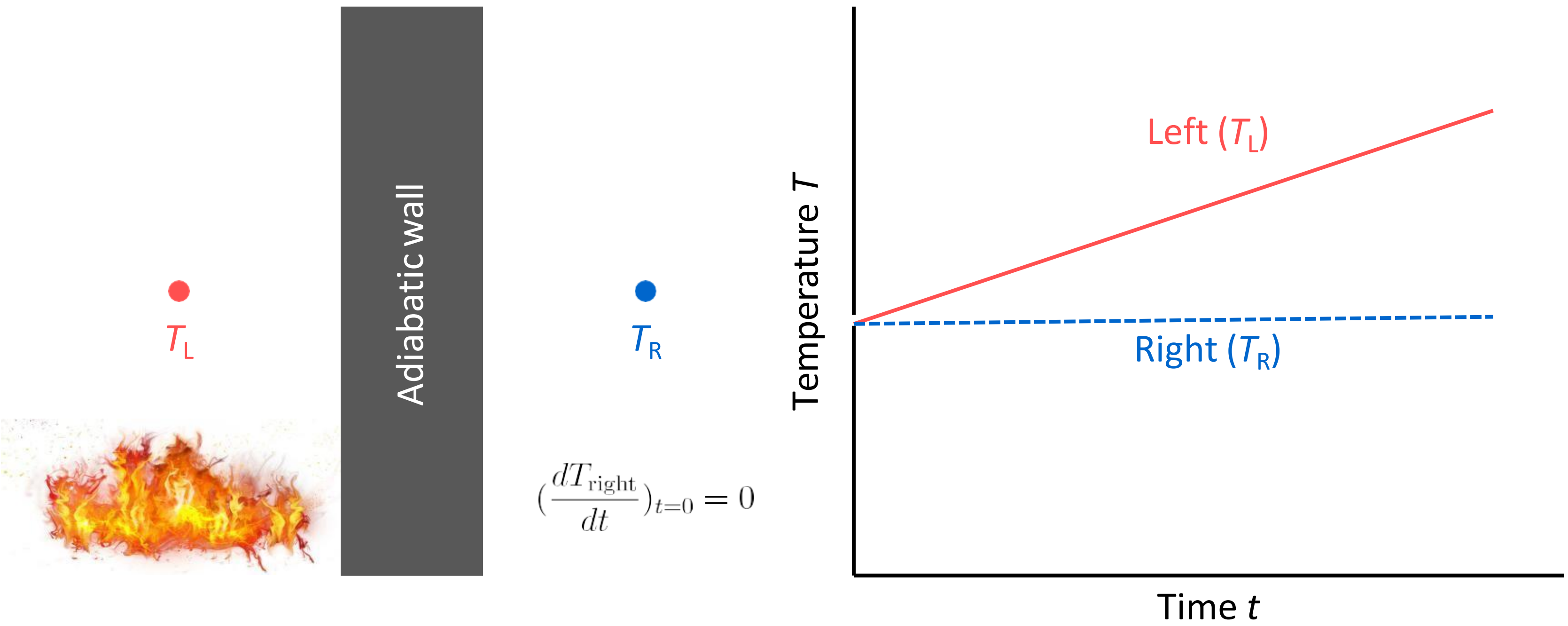
Operational Definition

❑ Adiabatic Wall

How do we operationally define an adiabatic wall without a prior definition of heat?

- **Operational Principle**

  Define an adiabatic wall purely through *observable temperature responses* on each body , without presupposing the unmeasured concept of heat



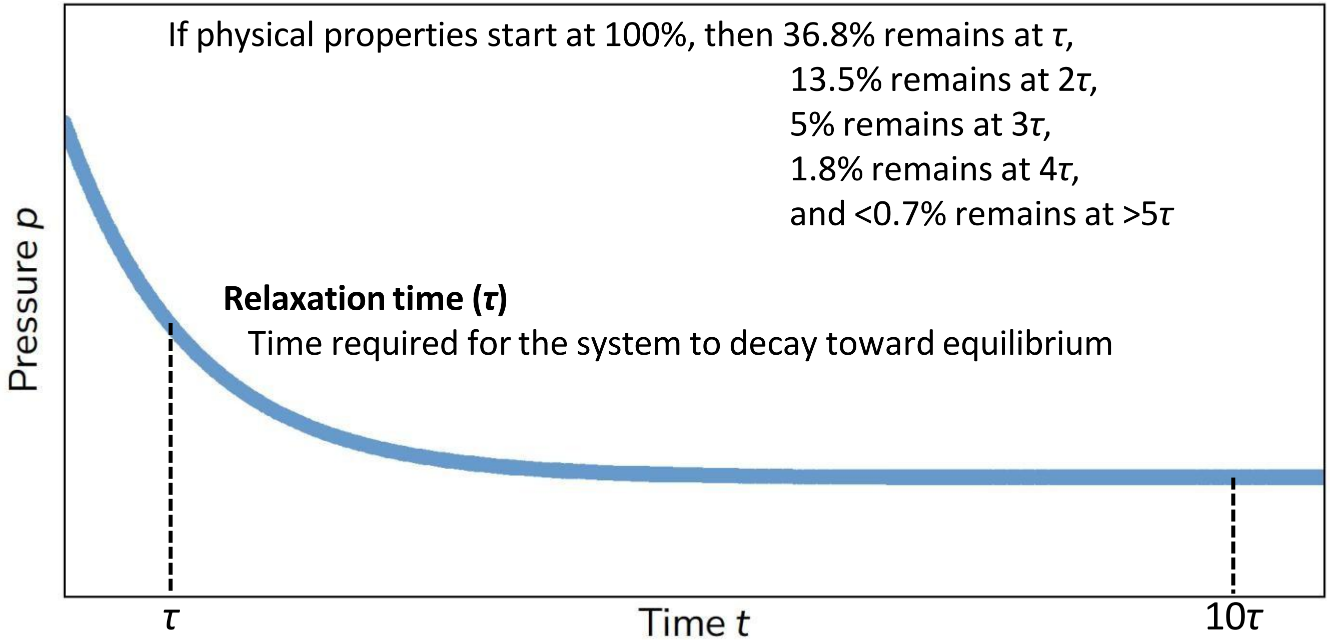
Operational Definition

❑ Thermodynamic Equilibrium

An isolated macroscopic system inevitably relaxes into a unique state of equilibrium

- **Equilibrium State**
  The final equilibrium state is completely independent of initial preparation
- **Relaxation Dynamics**
  Any macroscopic state variable evolves exponentially: $P(t) = P_\infty + (P_0 - P_\infty)\exp(-t/\tau)$
- **Operational Precision Criterion (The ε-Criterion)**
  Observation time must satisfy $\tau_{obs} \geq 10\tau \Leftrightarrow |P_\infty - P(10\tau)| = (P_0 - P_\infty)e^{-10} < \varepsilon$

  *$e^{-10} = 4.54\times10^{-5}$



Operational Definition

❑ Empirical Parameter (Heat Capacity)

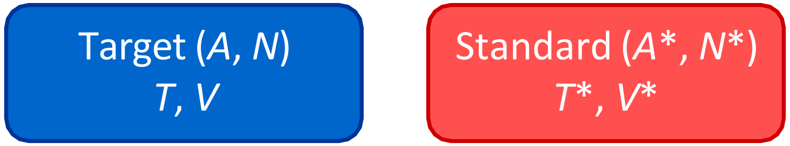
How do we operationally define an adiabatic wall without a prior definition of heat?

- **Operational Protocol**
  (1) Standard (e.g., water): $C = 1$ cal g^{-1} °C^{-1}
  (2) Diathermal contact
  (3) Linear empirical ratio (slope): $-\Delta T/\Delta T^*$

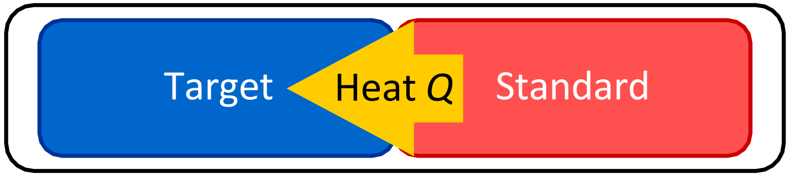
- **Operational Definition of Heat Capacity *C***
  Heat capacity is directly determined by the measured ratio a
  $C(T, V; A, N) = (1/a)C^*$

- **Operational Definition of Heat *Q***
  Heat is defined through measured temperature shifts, eliminating vague everyday intuition!
  $Q = C\Delta T = (1/a)C^* \times a\Delta T^* = -C^*\Delta T^*$

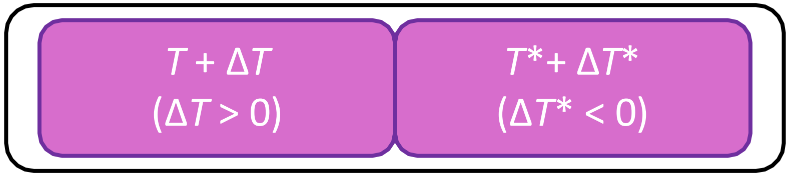
Prepared isolated bodies



Diathermal contact & heat flow



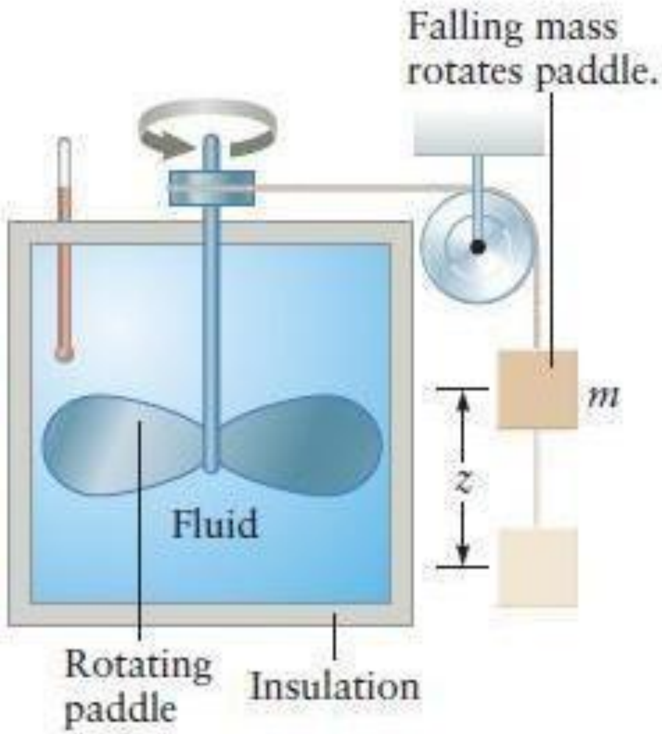
Equilibrium & Δ*T* measurement



Linear slope law: $\Delta T = -a\Delta T^*$ (as $\Delta T, \Delta T^* \to 0$)

Heat and Work

❑ Joule's Experiment



Apparatus used to measure the relation between heat energy and mechanical energy. See Insight

Insight

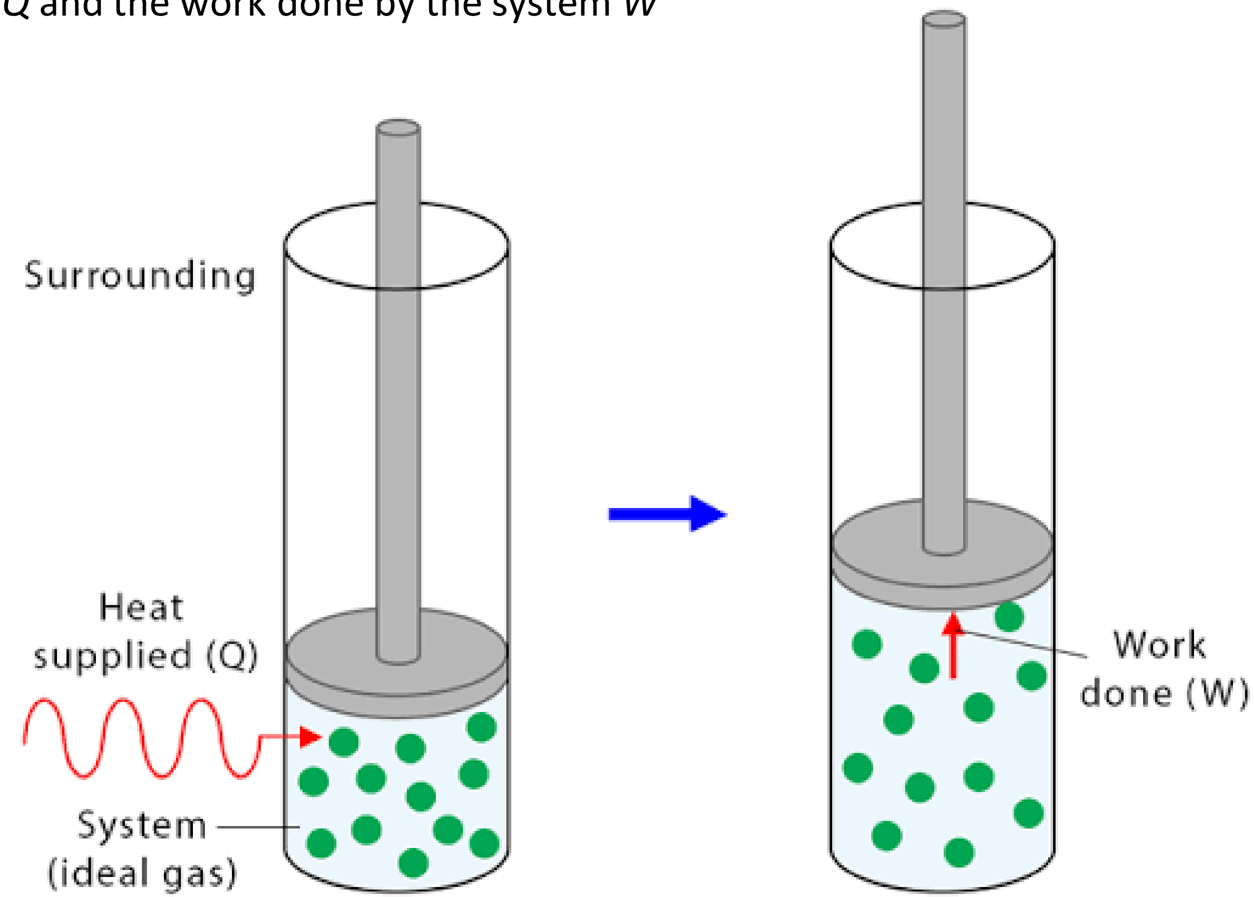
THE EQUIVALENCE OF MECHANICAL ENERGY AND HEAT ENERGY

In the 1840s, James Joule measured the "mechanical equivalent" of heat energy with an apparatus like the one shown in the figure . As a mass falls through a height z, its potential energy is used to rotate a propeller in a liquid. The propeller's motion heats the liquid, increasing the liquid's temperature. According to the principle of conservation of energy, the change in potential energy of the mass is converted to heat energy. In this way, Joule was able to relate the unit used at that time for heat energy (the calorie) to the unit of mechanical energy (the joule).

Heat and Work

❑ The First Law of Thermodynamics

The change in internal energy of a system ΔU equals the difference between the heat added to the system Q and the work done by the system W



$$\Delta U = Q - W$$

Heat and Work

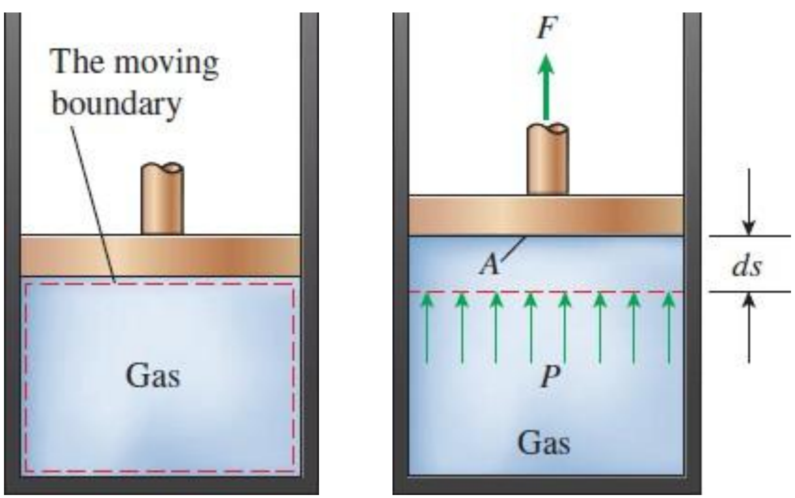
❑ Moving Boundary Work

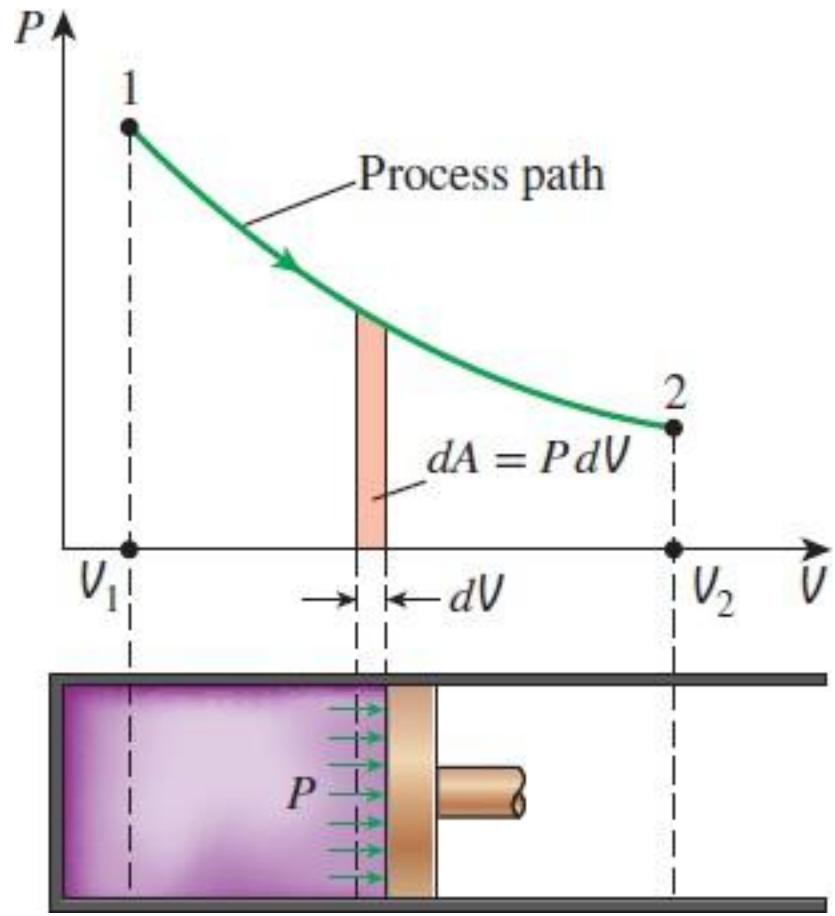
The work associated with moving boundary

(e.g.) Expansion and compression work
in a piston–cylinder device

When a piston under a force *F* moves a distance d*s*, the mechanical work *W* can be expressed as:

$$W = \int_{s_1}^{s_2} F ds = \int_{V_1}^{V_2} PA\frac{dV}{s} = \int_{V_1}^{V_2} PdV$$





Heat and Work

❑ Ideal Gas Law

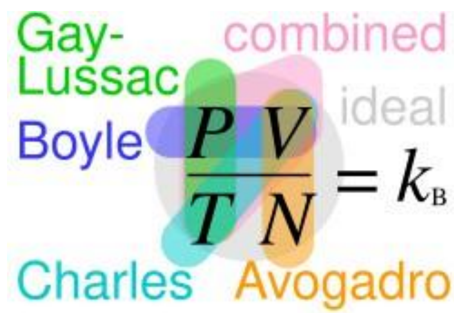
The equation of state that relates the pressure P, volume V, and temperature T for a given mass of an ideal gas, which has no repulsion and attraction between particles

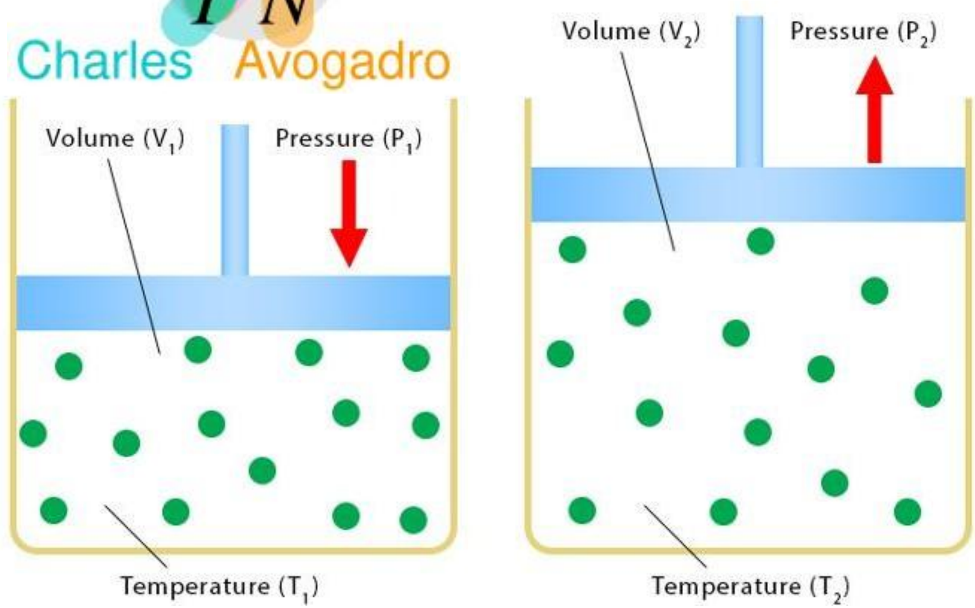
- **Boyle's Law (1662)**
  $P_1V_1 = P_2V_2$ (T = const.)

- **Charles' Law (1787)**
  $V_1/T_1 = V_2/T_2$ (P = const.)

- **Gay-Lussac's law (1808)**
  $P_1/T_1 = P_2/T_2$ (V = const.)

- **Avogadro's law (1811)**
  $V_1/n_1 = V_2/n_2$ (P & T = const.)





Combining the above laws,

$$PV = nRT = Nk_BT$$

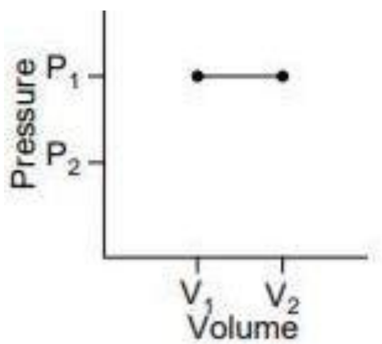
where n is the number of moles, R is the universal constant (8.31 J mol^{-1} K^{-1}),
N is the number of gas molecules and k_B is the Boltzmann constant (1.38×10^{-23} J K^{-1})

Heat and Work

❑ Thermodynamic Process

A change in the state of a system, defined by pressure, volume, and temperature, involving energy transfer as heat and work

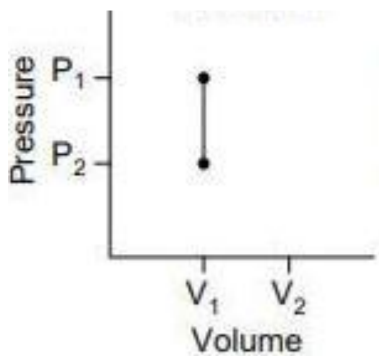
Isobaric ($dP = 0$)



$W = P_1(V_2 – V_1) = nR(T_2 – T_1)$
$Q = \Delta U + W = nC_P(T_2 – T_1)$

$C_P = (5/2)R$ (monoatomic)
$(7/2)R$ (diatomic)
$(9/2)\mathrm{R}$ (polyatomic)

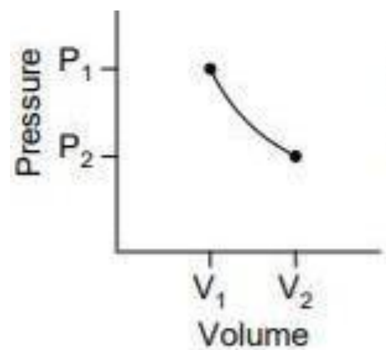
Isochroic ($dV = 0$)



$W = \int PdV = 0$
$Q = \Delta U = nC_V(T_2 – T_1)$

$C_V = (3/2)R$
$(5/2)R$
$(7/2)\mathrm{R}$

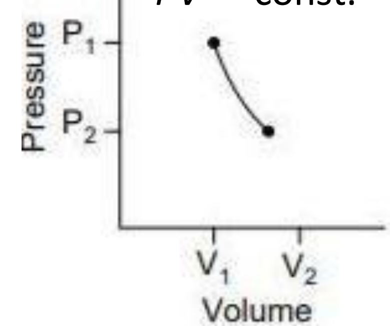
Isothermal ($dT = 0$)



$W = nRT_1\ln(V_2/V_1)$
$Q = W\ (\because \Delta U = 0)$

Adiabatic ($\delta Q = 0$)

PV^γ = const.



$W = (P_2V_2 – P_1V_1)/(1 – \gamma)$
$Q = 0$

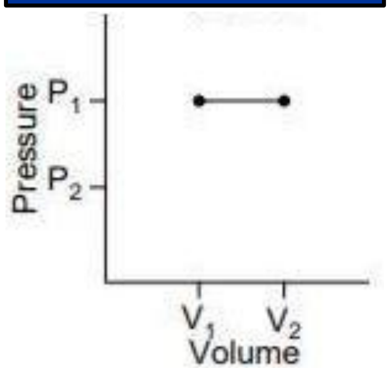
$\gamma = C_P/C_V = 5/3$
$7/5$
$9/7$

Heat and Work

❑ Thermodynamic Process

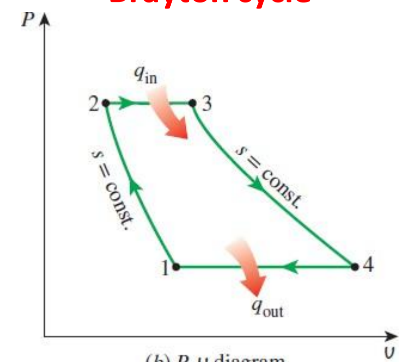
A change in the state of a system, defined by pressure, volume, and temperature, involving energy transfer as heat and work

Isobaric
($dP = 0$)



- **Specific steps in Brayton and Rankine cycles**
- **Heating water in an open pot on a stove**

Brayton cycle



(e.g.) 2 m³ volume of an ideal gas is stored under a constant pressure of 4×10^5 Pa. The temperature is increased till it expands to triple its initial volume. What is the work done?

$P_1 = 4 \times 10^5$ Pa
$V_1 = 2$ m³
$T_2 = 3T_1$
$\Rightarrow V_2 = \quad$ m³

$W = P_1(V_2 - V_1) = nR(T_2 - T_1)$
$Q = \Delta U + W = nC_P(T_2 - T_1)$

$C_P = (5/2)R$ (monoatomic)
$(7/2)R$ (diatomic)
$(9/2)R$ (polyatomic)

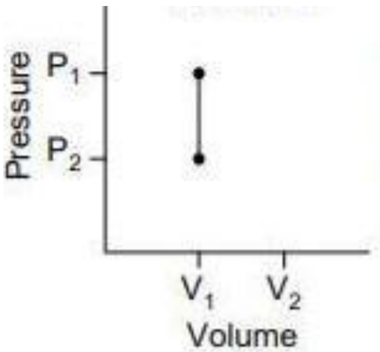
$\therefore W = P_1(V_2 - V_1) = 4 \times 10^5 \text{ Pa} \times (\quad -2) \text{ m}^3 = \qquad\qquad$ J

Heat and Work

❑ Thermodynamic Process

A change in the state of a system, defined by pressure, volume, and temperature, involving energy transfer as heat and work

Isochroic
($dV = 0$)

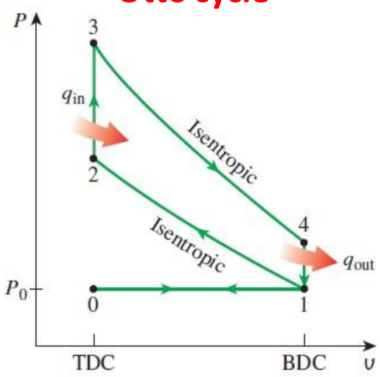


$W = \int PdV = 0$
$Q = \Delta U = nC_V(T_2 - T_1)$

$C_V = (3/2)R$ (monoatomic)
$(5/2)R$ (diatomic)
$(7/2)R$ (polyatomic)

- **The ideal Otto cycle used in automobile engines**
- **Tire pressure on a cold day**

Otto cycle



(e.g.) 2 kg of water is heated from 50 to 80 °C in a closed vessel. What will be the change in the internal energy of the water? (C_{water} = 4.2 J g^{-1} K^{-1})

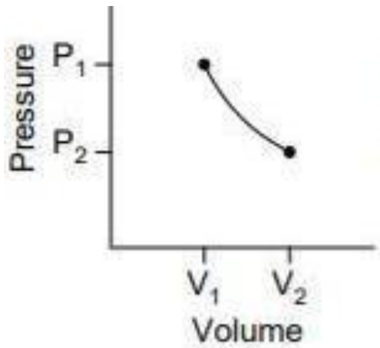
$\Delta V = 0 \Leftrightarrow W = 0$
$\therefore Q = \Delta U$
$\Rightarrow Q = mC_{water}\Delta T = 2\ \text{kg} \times 4.2\ \text{J g}^{-1}\ \text{K}^{-1} \times (80–50)\ \text{K} =$ kJ

Heat and Work

❑ Thermodynamic Process

A change in the state of a system, defined by pressure, volume, and temperature, involving energy transfer as heat and work

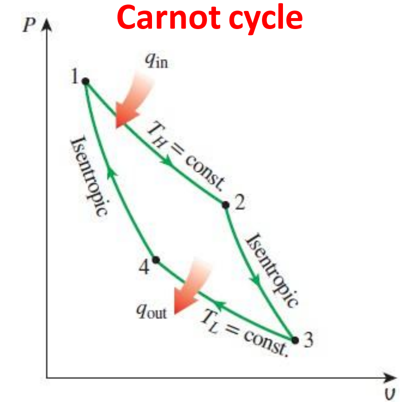
Isothermal
($dT = 0$)



$W = nRT_1\ln(V_2/V_1)$
$Q = W$ ($\because \Delta U = 0$)

- **Some parts of the Carnot engine**
- **Phase transformation occurs at constant temperatures (Melting, Boiling, Sublimation)**

Carnot cycle



(e.g.) A balloon contains 2 mol of air. It is kept at a constant temperature of 27 °C. The balloon contracts to 36.8% of its initial volume. Find the work done on the gas in contracting the balloon. (cf. e^{-1} = ~0.368)

n = 2 mol
T = 27 °C = 300 K
$V_2 = 0.368V_1$

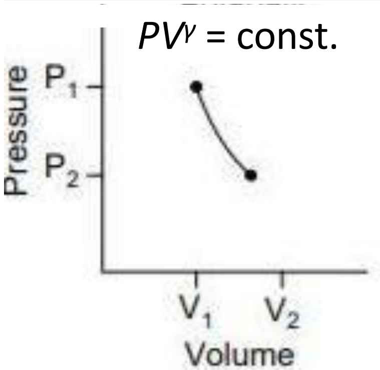
$W = nRT_1\ln(V_2/V_1)$
$= 2 \text{ mol} \times 8.31 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln(0.368) = \quad\quad \text{J}$

Heat and Work

Thermodynamic Process

A change in the state of a system, defined by pressure, volume, and temperature, involving energy transfer as heat and work

Adiabatic ($\delta Q = 0$)

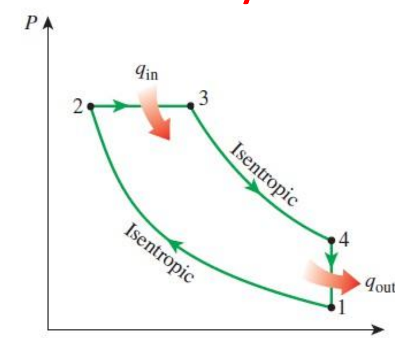


$W = (P_2V_2 - P_1V_1)/(1-\gamma)$
$Q = 0$

$\gamma = C_P/C_V$ = 5/3 (monoatomic)
7/5 (diatomic)
9/7 (polyatomic)

- **Certain parts of the Carnot and Diesel engines**
- **Rising & sinking air**
- **Blowing on hands (compression)**
- **Butane gas burner (compression)**

Diesel cycle



(a) P-υ diagram

(e.g.) Consider pumping air inside the tire of a bicycle using a hand pump. The air inside the pump is a thermodynamic system with well-defined volume, pressure, and temperature. Suppose the nozzle of the tire is blocked. When air is pumped, its volume reduces by 1/3. Calculate the final temperature of the air in the pump if the pump is at 27 °C initially.

T_1 = 27 °C = 300 K
$V_2 = (1/3)V_1$
$\gamma = C_P/C_V$ = 1.4 (since air is diatomic)

$T_2 = (V_1/V_2)^{\gamma-1}T_1 = 3^{0.4} \times 300$ K =   K

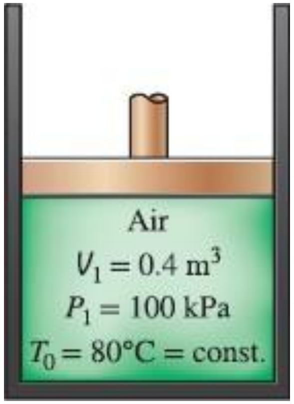
Heat and Work

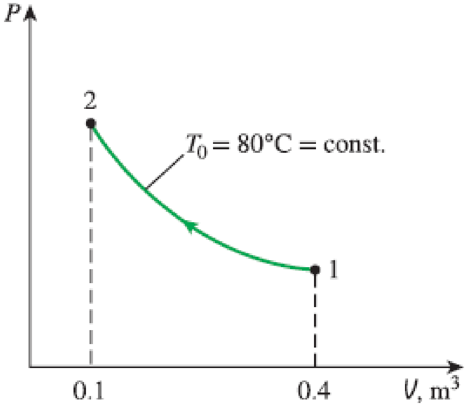
❑ Quiz

An ideal gas in a piston–cylinder device undergoes an isothermal process with an initial pressure and volume of P_1 and V_1, respectively.

- **Derive the work done by the gas W when it expands/compresses to a final volume V_2.**

- **A piston–cylinder device initially contains 0.4 m³ of air at 100 kPa and 80 °C. The air is now compressed to 0.1 m³ in such a way that the temperature inside the cylinder remains constant. Using the formula derived above, determine the work done during this process.**

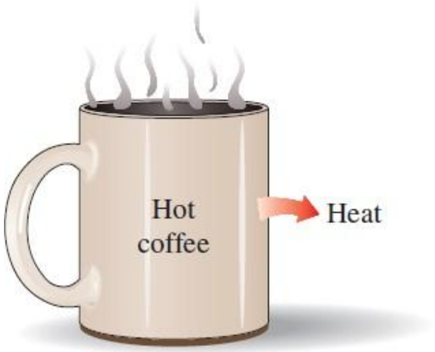




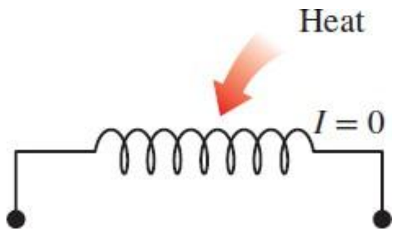
Entropy

❑ Limitation of The First Law of Thermodynamics

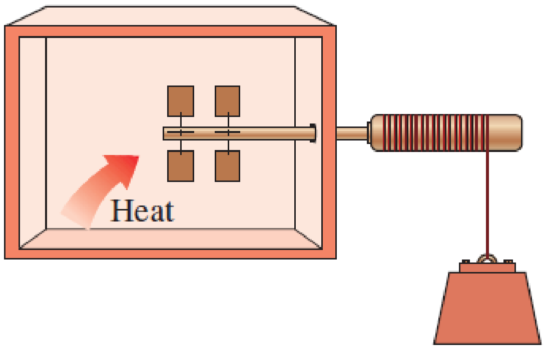
The first law of thermodynamics fails to predict *the direction of a process*, whether a reaction is spontaneous, or the limits of converting heat into work



A cup of hot coffee does not get in a cooler room



Transferring heat to a wire will not generate electricity



Transferring heat to a paddle wheel will not cause it to rotate

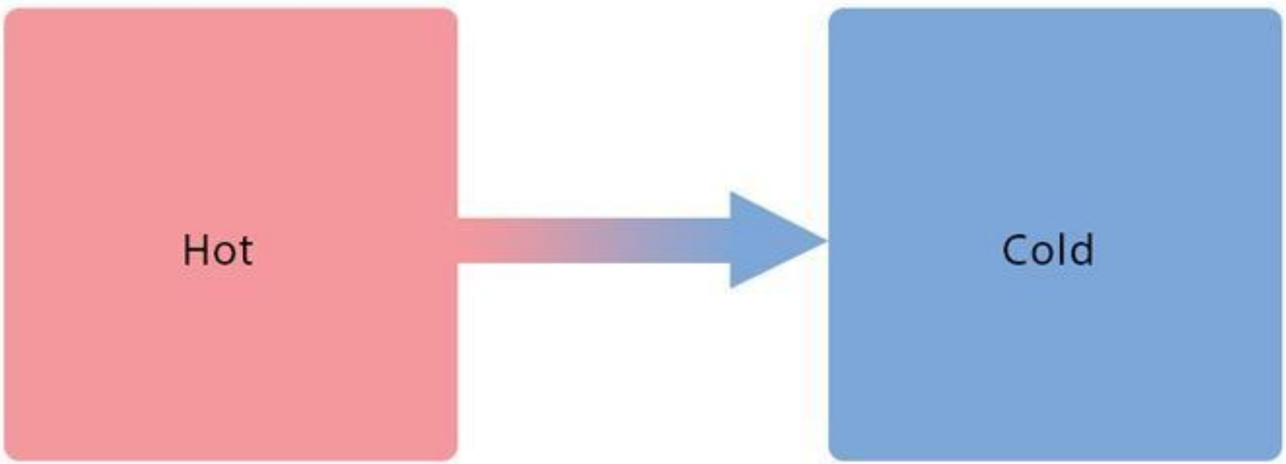
Entropy

❑ The Second Law of Thermodynamics

The entropy *S* of any natural and spontaneous process either increases or remains constant
(Not all heat can be converted into work in a cyclic process)

- **In a reversible, idealized process in a closed system** $dS = \frac{\delta Q}{T}$
- **From a perspective of statistical mechanics** $S = k_B \ln \Omega$

Example: Heat flow from a hot body to a cold body



$\boxed{\Delta S = 0}$ For reversible process

$\boxed{\Delta S > 0}$ For irreversible process

Entropy

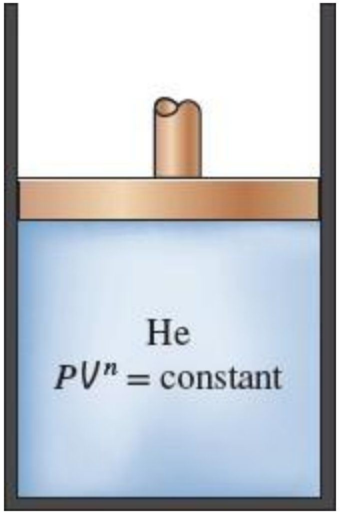
Reversible vs. Irreversible

Consider an insulated piston–cylinder device containing a monatomic ideal gas initially at 2 atm, 10 L, and 300 K. The surrounding atmospheric pressure is constant at 1 atm. Because the internal pressure is higher, the gas expands rapidly against the constant external pressure until the internal pressure reaches mechanical equilibrium at 1 atm. Determine the change in entropy of the gas.

P_1 = 2 atm, V_1 = 10 L, T_1 = 300 K

P_2 = 1 atm, V_2 = ?? L, T_2 = ??? K

$\gamma = C_P/C_V = (5/2)/(3/2) = 5/3$



	Reversible Process (Adiabatic)
Governing Eq.	PV^{γ} = const.
Final Vol. (V_2)	$V_2 = V_1(P_1/P_2)^{(1/\gamma)}$ $= 10\times(2/1)^{(3/5)}$ = **15.2 L**
Final Temp. (T_2)	$T_2 = T_1(P_2/P_1)^{(\gamma-1)/\gamma}$ $= 300\times(1/2)^{(2/5)}$ = **227.4 K**
Work (W)	$W_{rev} = (P_2V_2-P_1V_1)/(1-\gamma)$ $= (1\times15.2-2\times10)$ atm L $/(1-5/3)$ = **735 J**
Entropy Change (ΔS)	$\boldsymbol{\Delta S_{rev} = 0}$ **J K^{-1}**

Entropy

Reversible vs. Irreversible

Consider an insulated piston–cylinder device containing a monatomic ideal gas initially at 2 atm, 10 L, and 300 K. The surrounding atmospheric pressure is constant at 1 atm. Because the internal pressure is higher, the gas expands rapidly against the constant external pressure until the internal pressure reaches mechanical equilibrium at 1 atm. Determine the change in entropy of the gas.

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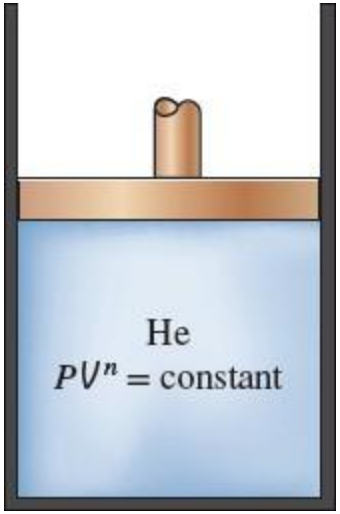
P_2 = 1 atm, V_2 = ?? L, T_2 = ??? K

$\gamma = C_P/C_V = (5/2)/(3/2) = 5/3$



	Irreversible Process (Adiabatic)
Governing Eq.	$\Delta U = -W \Leftrightarrow nC_V(T_2-T_1) = -P_{ex}(V_2-V_1)$ where $P_{ex} = P_2 = nRT_2/V_2$
Final Vol. (V_2)	$1.5nR(T_2-T_1) = 1.5(P_2V_2-P_1V_1) = -P_2V_2+P_2V_1$ $\Leftrightarrow V_2 = V_1[0.6(P_1/P_2)+0.4P_2]$ $= 10 \times [0.6\times(2/1)+0.4\times1]$ $=$ **16 L**
Final Temp. (T_2)	$T_2 = P_2V_2/nR = T_1(P_2V_2/(P_1V_1)$ $= 300\times[(1\times16)/(2\times10)]$ $=$ **240 K**
Work (W)	$W_{irrev} = P_{ex}(V_2-V_1)$ $= 1\times(16-10)$ atm L $=$ **608 J (<735 J = W_{rev})**

Entropy

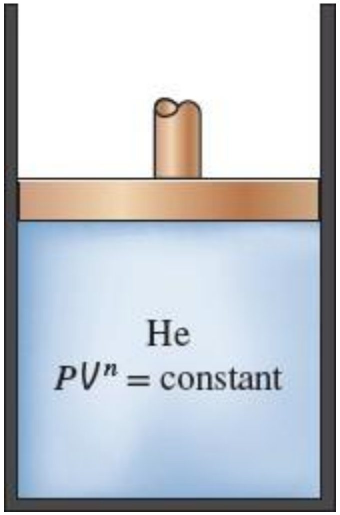
Reversible vs. Irreversible

Consider an insulated piston–cylinder device containing a monatomic ideal gas initially at 2 atm, 10 L, and 300 K. The surrounding atmospheric pressure is constant at 1 atm. Because the internal pressure is higher, the gas expands rapidly against the constant external pressure until the internal pressure reaches mechanical equilibrium at 1 atm. Determine the change in entropy of the gas.

P_1 = 2 atm, V_1 = 10 L, T_1 = 300 K

P_2 = 1 atm, V_2 = ?? L, T_2 = ??? K

$\gamma = C_P/C_V = (5/2)/(3/2) = 5/3$



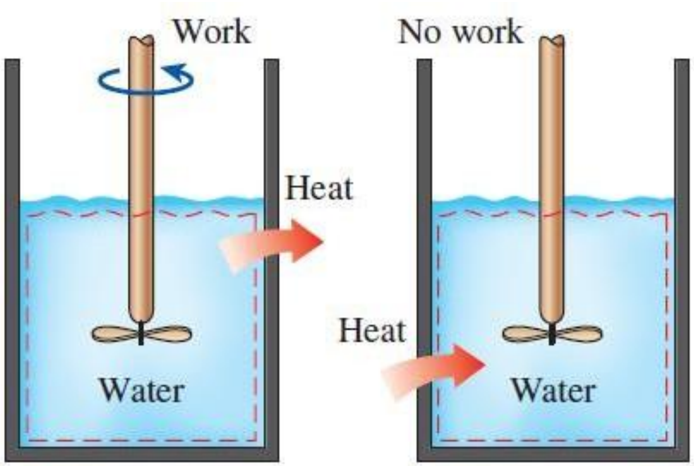
	Irreversible Process (Adiabatic)
Path (since ΔS is a state function)	(1) Isothermal $(P, V, T) = (2, 10, 300) \rightarrow (1.25, 16, 300)$ (2) Isochoric $(P, V, T) = (1.25, 16, 300) \rightarrow (1, 16, 240)$
Entropy Change (ΔS)	$\Delta S_{\mathrm{irrev},1} = nRT_1\ln(V_2/V_1) / T_1$ $= (2\times10)\ln(16/10) / 300$ atm L $= \mathbf{3.18\ J\ K^{-1}}$ $\Delta S_{\mathrm{irrev},2} = \int dQ/T = \int dU/T = \int_{T1}^{T2} nC_V dT / T$ $= nC_V\ln(T_2/T_1)$ $= 1.5\times1\times16/240\ \ln(240/300)$ atm L $= \mathbf{-2.26\ J\ K^{-1}}$ $\therefore \boldsymbol{\Delta S_{\mathrm{irrev}}} = \Delta S_{\mathrm{irrev},1}+\Delta S_{\mathrm{irrev},2} = \Delta S_{\mathrm{system}}$ $\mathbf{= 0.92\ J\ K^{-1} > \Delta S_{rev}}$

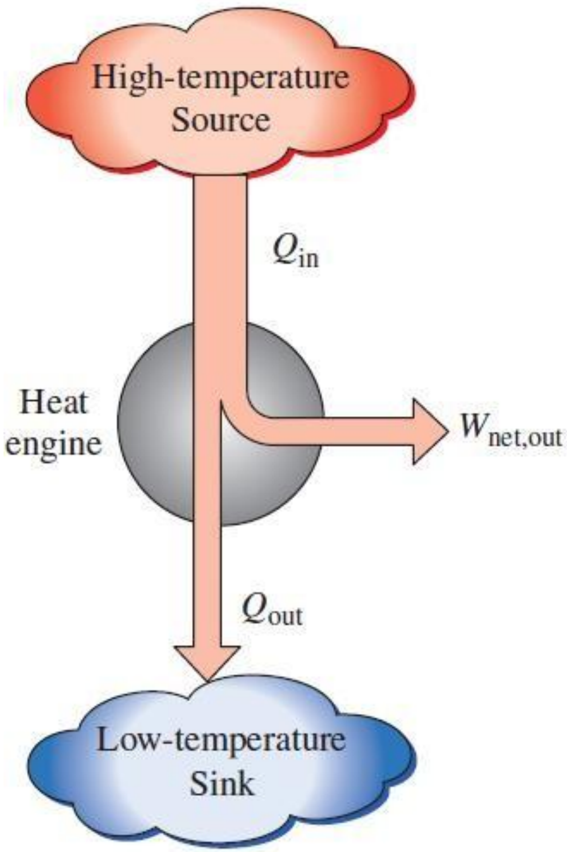
Entropy

❑ Heat Engines

Mechanical work can be converted to heat directly and completely, but converting heat to mechanical work requires the use of some special devices, called *heat engines*

- **Heat engines receive heat from a high-temp. source (e.g., solar energy, furnace, nuclear reactor, etc.)**
- **They convert part of this heat to mechanical work (usually in the form of a rotating shaft)**
- **They reject the remaining waste heat to a low-temp. sink (atmosphere, river, etc.)**
- **They operate on a cycle**





$$\eta_{th} = 1 - \frac{Q_L}{Q_H}$$

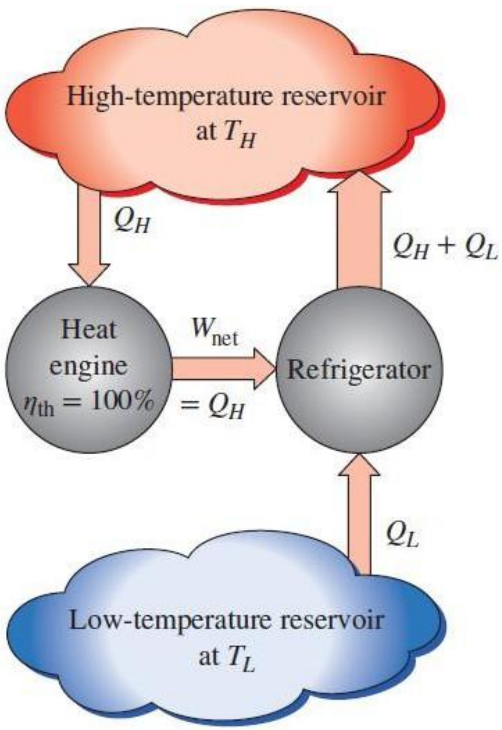
Entropy

❑ Kelvin–Planck Statement

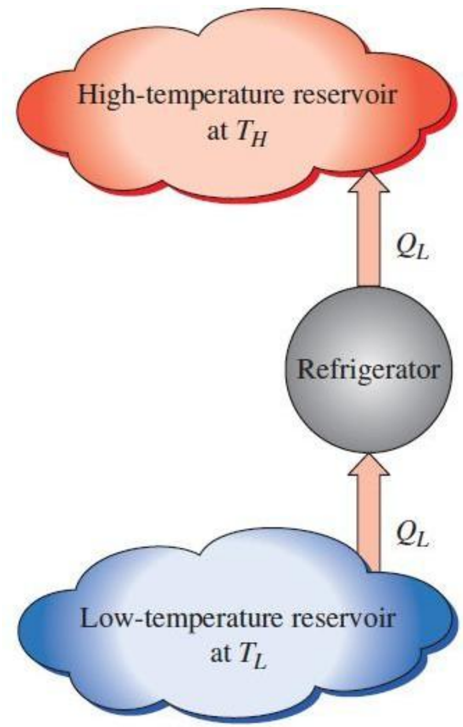
It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work

❑ Clausius Statement

Heat can never pass from a colder to a warmer body without some other change occurring at the same time



(a) A refrigerator that is powered by a 100 percent efficient heat engine



(b) The equivalent refrigerator

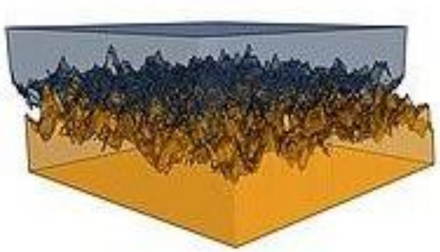
Entropy

❑ Irreversibility

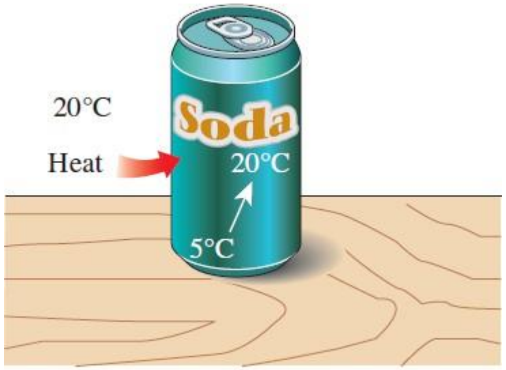
The factors that cause a process to be irreversible

- **Death**

- **Friction**



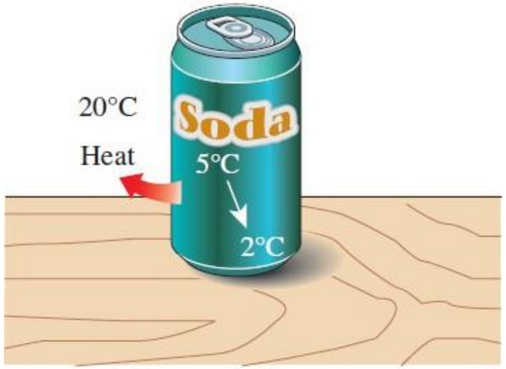
- **Heat Transfer**



(*a*) An irreversible heat transfer process

- **Time**

- **Free Expansion**



(*b*) An impossible heat transfer process

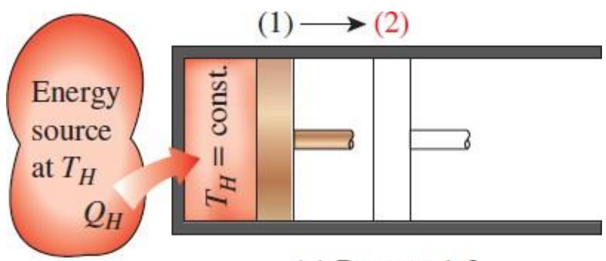
Entropy

❑ Carnot's Theorem (1824)

All heat engines operating between the same two thermal or heat reservoirs cannot have efficiencies greater than a reversible heat engine operating between the same reservoirs

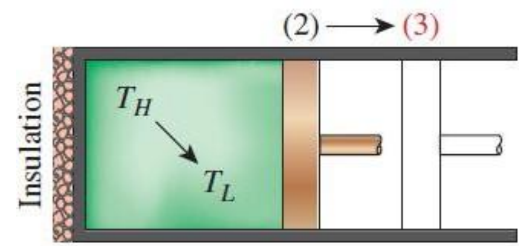
Carnot Engine (Ideal Heat Engine) $\eta_{\text{th}} = 1 - \dfrac{Q_L}{Q_H} = 1 - \dfrac{T_L}{T_H}$

(a) Reversible isothermal expansion



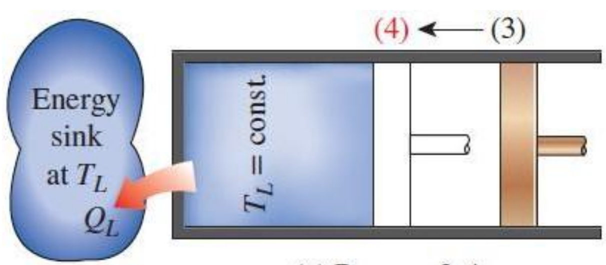
(a) Process 1-2

(b) Reversible adiabatic expansion



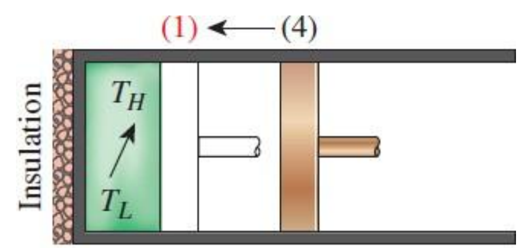
(b) Process 2-3

(c) Reversible isothermal compression



(c) Process 3-4

(d) Reversible adiabatic compression



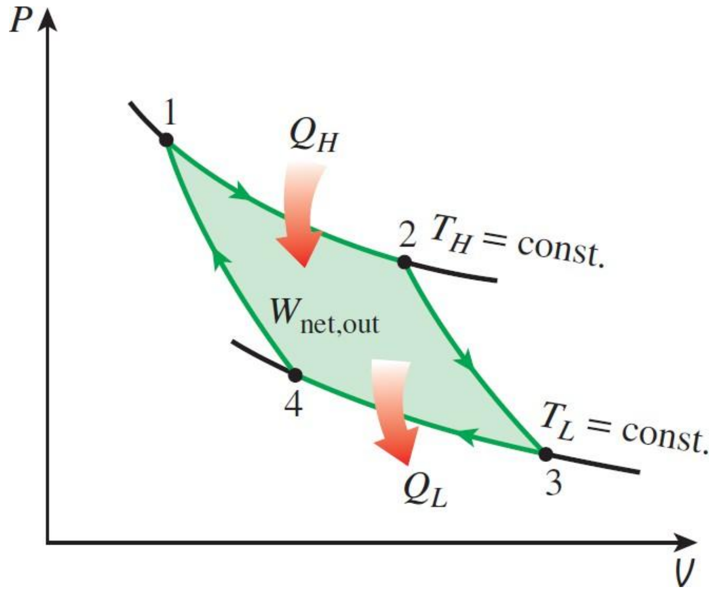
(d) Process 4-1

Entropy

❑ Carnot's Theorem (1824)

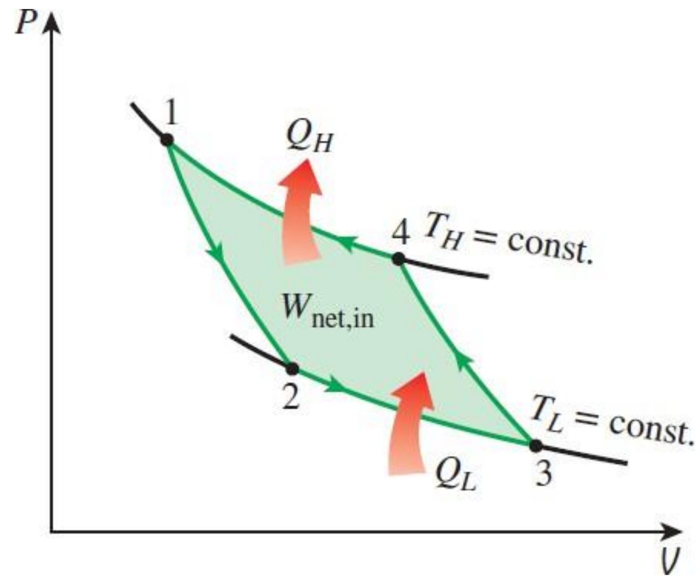
All heat engines operating between the same two thermal or heat reservoirs cannot have efficiencies greater than a reversible heat engine operating between the same reservoirs

***P–V* diagram of the Carnot cycle (Heat Engine)**



$$\eta = \frac{W}{Q_H} = 1 - \frac{T_L}{T_H} \quad (< 100\%)$$

***P–V* diagram of the reversed Carnot cycle (Refrigerator, Heat Pump)**



$$\mathrm{COP}_R = \frac{Q_L}{W_{\mathrm{in}}} = \frac{T_L}{T_H - T_L}$$

$$\mathrm{COP}_{HP} = \frac{Q_H}{W_{\mathrm{in}}} = \frac{T_H}{T_H - T_L} = \mathrm{COP}_R + 1$$

Assignment (~15th Sep.)

❑ Carnot's Theorem (1824)

All heat engines operating between the same two thermal or heat reservoirs cannot have efficiencies greater than a reversible heat engine operating between the same reservoirs

- **Express the heat absorbed by the gas from the high-temperature reservoir Q_H (process 1 to 2) and the magnitude of heat rejected to the low-temperature reservoir Q_L (process 3 to 4), in terms of n, R, T_H, T_L, and the corresponding volumes (V_1, V_2, V_3, V_4).**
  (※ Hint: Use the first law of thermodynamics and note that the internal energy of an ideal gas depends solely on temperature: $dU = nC_V dT$)

Assignment (~15th Sep.)

❑ Carnot's Theorem (1824)

All heat engines operating between the same two thermal or heat reservoirs cannot have efficiencies greater than a reversible heat engine operating between the same reservoirs

- **Using the relation for a reversible adiabatic process of an ideal gas $TV^{\gamma-1}$ = constant ($\gamma = C_P/C_V$), analyze the two adiabatic paths (processes 2 to 3 & 4 to 1) to prove that:**

$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$

- **Using the general definition of thermal efficiency for a heat engine ($\eta_{th} = 1 - Q_L/Q_H$), prove that the efficiency of a Carnot engine η_{Carnot} is become**

$$\eta_{\mathrm{Carnot}} = 1 - \frac{T_L}{T_H}$$

End of Document